Received 20 July 2022, accepted 6 August 2022, date of publication 8 August 2022, date of current version 16 August 2022.
Digital Object Identifier 10.1109/ACCESS.2022.3197629

TOPICAL REVIEW

Low-Light Image Enhancement: A Comparative Review and Prospects

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This work was supported by the National Research Foundation of Korea (NRF) Grant through the Korean Government [Ministry of Science and ICT (MSIT)] under Grant 2020R1F1A1068080.

ABSTRACT Low-light image enhancement is a key prerequisite for diverse applications in the field of image processing and computer vision. Various approaches for this task have been introduced over last few decades, and the current state of the art methods have shown remarkable advances based on deep neural networks. However, there are still technical issues to be resolved, e.g., dependency on subjective re-touching results and inconsistency with subjective evaluations. The goal of this work is to provide a comprehensive overview and a practical guide for experts as well as beginners. This paper covers a systematic taxonomy of existing algorithms, representative methodologies, and the performance analysis on benchmark datasets. To pave the way of the development direction for low-light image enhancement, constructive discussions and prospects are also provided.

INDEX TERMS Low-light image enhancement, deep neural networks, comprehensive overview, constructive discussions and prospects.

I. INTRODUCTION

With the rapid increase of camera-mounted devices, e.g., smartphones and tablets, various aspects of daily life have been easily recorded and shared by the huge amount of photos. In order to provide more visually-pleasing images to individuals, there have been notable technological advances for camera sensors in terms of hardware, however, diverse scenarios occurring in real-world environments still make unpredictable degradations particularly under low-lighting conditions. For example, occlusions often cast a shadow intricately structured onto the target object and backlights yield undesirable dark areas against the bright background (see Fig. 1). The purpose of low-light image enhancement is to improve the visibility without severe distortions in underlying contents of a given image. Most typically, several adjustment techniques based on the domain knowledge and statistical properties (e.g., Gamma correction and histogram equalization [1]) have been widely employed for this task and shown satisfactory results for globally-degraded low-light images. However, such approaches do not cover various situations occurring in real-world environments, which make more complicated (i.e., spatially uneven and nonlinear) dark areas in captured images. Furthermore, more challenging issues to utilize the technique for low-light image enhancement as an essential preprocessing module still remain, which can be briefly summarized as follows:

- *Ill-posedness Nature:* it is not possible to know the properties of the currently captured low-light images under normal lighting conditions, thus the answer to the question whether the visual quality of the enhancement result is the best or not can be varied according to users. Moreover, since it is intractable to separate illumination components from a single image, enhancement results are probably suboptimal.
- *Learning Bias:* most state-of-the art methods adopt the supervised learning scheme based on deep network architectures, however, the ground truth for benchmark datasets often has been produced by a few experts, which tends to guide the learning procedure based on the subjective criteria. Even though paired low/normal light images are collected through altering the exposure time and ISO during the acquisition process for dataset

The associate editor coordinating the review of this manuscript and approving it for publication was G. R. Sinha.

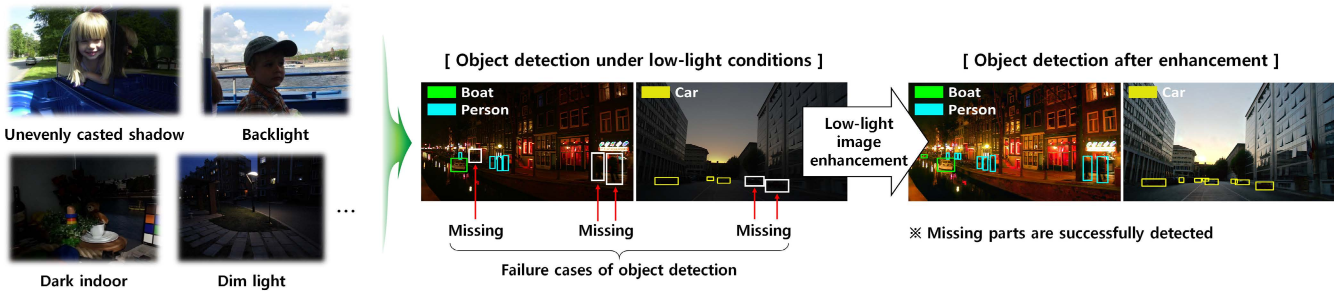

FIGURE 1. Examples of low-light conditions and the effect of low-light image enhancement for object detection. Note that the object detector [6] often fails to grasp target objects under low-light conditions (white rectangles indicate some examples of missing parts). The enhancement in this example is conducted by DSLR [51].

construction, it does not sufficiently reflect environmental factors of real-world scenarios.

To tackle those challenging issues, various methods particularly focusing on deep learning techniques have been actively introduced to date. As an example, the no-reference learning scheme [2] has started to be developed for reduction of the learning bias as mentioned above. Note that detailed explanations will be provided in the following Sections.

The application scope of low-light image enhancement is quite wide and representatively includes media quality improvement, scene understanding for autonomous driving, intelligent surveillance, high-performance security, etc. Specifically, low-light conditions give significant difficulties to user authentication on mobile devices since captured facial images are probably different from enrolled ones. Therefore, low-light image enhancement can play an important role to construct the reliable verification system by improving the visibility of facial images. In the scenario of autonomous driving, unevenly-casted shadows often yield irregular dark areas on lanes, thus adjustment of such challenging conditions guides the algorithms for scene understanding, e.g., object detection and recognition in road environments, to be more reliable. For safety and security of the vessel traffic in maritime applications, it is essential to guarantee reliable vessel detection under low-visibility conditions via low-light image enhancement [3]. Based on these practical examples, it is naturally considered that the demand for the deployment of low-light image enhancement into various industrial fields keeps growing year after year. In line with this need, the considerable research has been done not only by academic institutions, but also by private companies.

This paper provides a comprehensive review of low-light image enhancement with extensive experimental results. Even though several surveys already appeared in literature [4], [5], to the best of our knowledge, this is the first work to give a in-depth overview between handcrafted feature-based and learning-based approaches in a balanced way. In contrast to previous surveys, we conduct a review focusing on representative methods rather than including all traditional methods that are somewhat outdated while paying attentions on deep learning-based methods introduced in the most recent literature. Moreover, strengths and drawbacks of each category are systematically analyzed with extensive experimental results. Through providing both qualitative and quantitative results of representative methods for low-light image enhancement, this paper serves as a practical guide for beginners to successfully contribute to this topic. Finally, the future direction of low-light image enhancement models is provided. The main contributions of this paper can be summarized as follows:

- This review handles core aspects in the problem of low-light image enhancement, e.g., how to formulate the enhancement process, how to effectively categorize existing methods for this task, what should be considered to generate the reliable enhancement results, etc. All the methods are reviewed and analyzed in line with this point of view. It is believed that answers to these constructive questions inspire many researchers to contribute to build the reliable system for low-light image enhancement.
- In contrast to previous comparative studies concentrating only on one side, e.g., handcrafted feature-based methods [4] or deep learning-based methods [5], this review takes into account various categories in a balanced way based on representative methods rather than explaining all the outdated methods. To help readers understand clearly, we also provide the systematic taxonomy defined according to types of models and analysis tools. This is fairly desirable for newcomers to compactly and effectively understand the research trend of low-light image enhancement. Moreover, discussions for pros and cons of each category are expected to give a great help to understand why the research direction is progressing in the current way.
- Instead of providing the performance comparison only with a few images [4], experimental results of representative methods in each category are sufficiently provided both qualitatively and quantitatively for various benchmark datasets. Based on the detailed analysis of performance evaluations, meaningful prospects as well as the general conclusion for the future of low-light image enhancement are also drawn.

TABLE 1. Taxonomy for low-light image enhancement. Note that representative methods belonging to each category are grouped according to key features rather than simply referring to individual algorithms.

Reference	Category	Model	Analysis	Key features of methodologies
Reza [10]	Handcrafted	Statistic.	Histogram	* Local histogram equalization
Sun *et al.* [11]	Handcrafted	Statistic.	Histogram	* Histogram specification
Menotti *et al.* [12]	Handcrafted	Statistic.	Histogram	* Multi-histogram equalization
Ueda & Suetake [13]	Handcrafted	Statistic.	Histogram	* Hue-preserving color histogram modification
Celik & Tjahjadi [14]	Handcrafted	Statistic.	Histogram	* 2D histogram modification
Lee *et al.* [15]	Handcrafted	Statistic.	Histogram	* 2D histogram modification
Gu *et al.* [16]	Handcrafted	Statistic.	Histogram	* Saliency-guided histogram equalization
Gu *et al.* [18], [66]	Handcrafted	Statistic.	Histogram	* Histogram combined with the result of image quality assessments
Kimmel *et al.* [19]	Handcrafted	Decomp.	Optimization	* Illumination prior-based standard framework
Fu *et al.* [21]	Handcrafted	Decomp.	Optimization	* A weighted variational framework
Guo *et al.* [22]	Handcrafted	Decomp.	Optimization	* Sequential optimization framework
Li *et al.* [23]	Handcrafted	Decomp.	Optimization	* Standard decomposition with direction minimization technique
Ren *et al.* [24]	Handcrafted	Decomp.	Optimization	* Low-rank regularized decomposition framework
Hao *et al.* [25]	Handcrafted	Decomp.	Optimization	* Semi-decouple decomposition framework
Jobson *et al.* [26], [27]	Handcrafted	Decomp.	Descriptor	* Gaussian residual in the log space (single-scale and multi-scale)
Wang *et al.* [28]	Handcrafted	Decomp.	Descriptor	* Maximum of RGB as the initial illumination
Liang *et al.* [29]	Handcrafted	Decomp.	Descriptor	* Nonlinear diffusion filtering
Kim [31], [36]	Handcrafted	Decomp.	Descriptor	* Orthogonal transform-based illumination descriptor
Ignatov *et al.* [37]	Learning	Ref.	DNN(sup.)	* Simple stacked convolutional layers with various loss functions
Lv *et al.* [38]	Learning	Ref.	DNN(sup.)	* Simple U-shaped convolutional architecture
Wang *et al.* [44]	Learning	Ref.	DNN(sup.)	* Decomposition-based staged architecture
Zhang *et al.* [45]	Learning	Ref.	DNN(sup.)	* Decomposition-based staged architecture
Lim & Kim [51]	Learning	Ref.	DNN(sup.)	* Multi-scale feature analysis-based network framework
Park *et al.* [54]	Learning	No-ref.	DNN(unsup.)	* Reinforcement learning scheme
Guo *et al.* [2]	Learning	No-ref.	DNN(unsup.)	* Estimation of the mapping curve for low-normal light pairs
Yang *et al.* [39]	Learning	No-ref.	DNN(unsup.)	* Comparison via re-generated unpaired normal images
Jiang *et al.* [58]	Learning	No-ref.	DNN(unsup.)	* GAN-based loss for the consistency of underlying properties

The rest of this paper is organized as follows. The overall procedure of low-light image enhancement and the systematic taxonomy are presented in Section II. The representative methods for each category are introduced in detail in Section III. The performance comparison based on benchmark datasets is demonstrated and the corresponding discussion is also presented in Section IV. Conclusions follow in Section V.

II. CONCEPT AND SYSTEMATIC TAXONOMY

A. GENERAL FORMULATION OF LOW-LIGHT IMAGE ENHANCEMENT

Over last few decades, diverse methods for low-light image enhancement have been introduced in the field of computer vision. Even though their philosophies for designing the algorithm are quite different each other, most previous methods share a common assumption that intrinsic properties of underlying contents (e.g., reflectance of objects, background color, noise level, etc.) should be preserved during the enhancement process. The overall procedure of low-light image enhancement can be simply formulated as follows:

$$E(x, y) = \mathcal{F}(f(x, y)), \tag{1}$$

where $f(x, y)$ and $E(x, y)$ denote the pixel value at the position (x, y) of the input low-light image and the corresponding enhancement result, respectively. $\mathcal{F}(\cdot)$ is the function representing the enhancement process, which is not limited to the specific type. For example, various optimization tools and deep neural networks can be adopted for describing such nonlinear and complicated relationship between input and output. Therefore, it is naturally thought that the difference between existing methods stems from the strategy to define $\mathcal{F}(\cdot)$ in (1). Since the ground truth is not given in most cases of real-world scenarios, each method aims at generating the reliable enhancement result, i.e., $E(x, y)$, based on well-established metrics that allow for subjective as well as objective criteria appropriately. This review digs into such strategies of each method to help readers better understand based on the systematic categorization, which will be explained in the following subsection.

B. SYSTEMATIC TAXONOMY

In general, it is difficult to define a single unambiguous criteria for classifying numerous low-light image enhancement methods. Nevertheless, existing methods can be broadly divided into two major groups, i.e., handcrafted feature-based

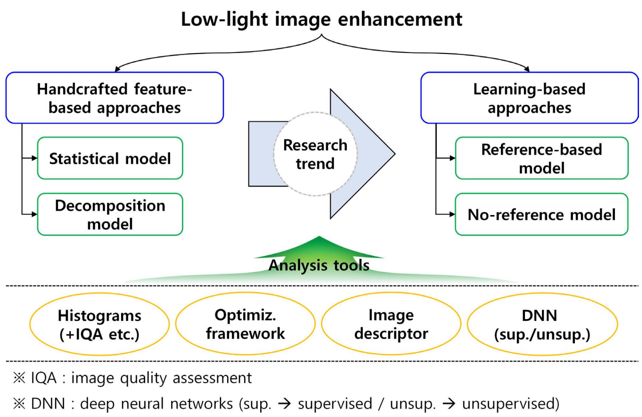

FIGURE 2. Systematic taxonomy for low-light image enhancement methods. Note that the research trend has moved from handcrafted feature-based approaches to learning-based approaches.

approaches and learning-based approaches. The former is further classified into two sub-models, i.e., statistical and decomposition models, while the latter consists of reference-based and no-reference models, which will be explained in detail in the following Section. In terms of timeline, the research trend has led from the handcrafted feature-based approach to the recent development of the learning-based approach. Figure 2 show this taxonomy in a hierarchical manner and corresponding details of each category with representative methods are shown in Table 1.

In the beginning, most researchers have attempted to explore and design various image features for low light image enhancement. Specifically, the statistical information of pixel intensities has been widely adopted to reach the desirable distribution in a global viewpoint. This strategy works well for the case of low contrast images (e.g., overall dark images), however, it often fails to take into account spatially-varying lighting effects, which yield the unbalanced intensity distribution in local regions. To successfully cover such real-world environments, many studies have concentrated on the principle that the pixel value is define by product of illumination and reflectance [1] (also known as the Retinex theory [7]). That is, by separating illumination components from a given image, lighting effects can be effectively adjusted and the corresponding result is subsequently re-combined to reflectance for generation of the enhanced image. To do this, various methods have been actively proposed in literature, which are based on mathematical engineering and optimization theories, while image descriptors also have been exploited to infer the illumination component in a closed-form solution. On the other hand, to be coherent with the subjective perception on enhancement results, several studies have applied the visual quality metric to estimate the enhancement process, i.e., $\mathcal{F}(\cdot)$ in (1). Those handcrafted feature-based approaches still have shown reliable results in a wide range of indoor and outdoor scenarios to date.

Inspired by the great success of deep learning-based approaches in the field of image classification and recognition, several researchers have started to applied the deep learning schemes to the problem of low-light image enhancement. As alternatives to handcrafted features, these approaches have explored the relationship between input (low-light) images and output (normal-light) images via learning techniques. To do this, the dataset, which is exquisitely designed based on paired images, is essential. Recently, most algorithms for low-light image enhancement have been actively introduced with a variety of neural network architectures. In order to elaborately guide the learning process, various loss functions also have been designed. Even though such reference-based learning schemes have shown the remarkable improvement without using any domain knowledge, building the reliable dataset still remains a challenge. For example, to make the ground truth (i.e., normal-light images), a few experts conduct the re-touching process with original low-light images [8]. However, this strategy gives a learning bias to the enhanced result by the subjective criteria of experts. The other dataset [9] is constructed based on changing the exposure time and ISO to cover a wide range of lighting conditions, however, it also has a limitation to consider diverse lighting effects under real-world environments. Most recently, to cope with this limitation, several studies have started to develop the unpaired learning scheme, which do not require producing the ground truth artificially. Furthermore, the zero-reference model, i.e., no need for paired or unpaired data, has been introduced with newly designed loss functions. Even though the performance of such new attempts is somewhat dropped compared to reference-based methods, many researchers believe that this is the right direction to go further to build the reliable enhancement system working well under unpredictable real-world environments.

Based on the taxonomy presented above, methodologies of diverse approaches for low-light image enhancement are explained in detail in the following Section.

III. METHODOLOGIES

A. HANDCRAFTED FEATURE-BASED APPROACHES

1) STATISTICAL MODELS

From the early days of the computer vision research, the statistical information over all pixel values have been most widely used for various applications. The histogram represents such global statistics simply and compactly in terms of the density of pixel values, thus many researchers adopted it to modify the distribution of pixel values for contrast enhancement, i.e., adjust the dynamic range of pixel intensities. The histogram equalization [1] is the most representative method and has been consistently employed as baseline for this task. To cover illumination variations occurring in local regions more efficiently, the block-based histogram equalization technique was developed with the clipping constraint [10] and has shown reliable enhancement results for medical as well as natural images. Inspired by the outstanding performance despite simple operations, various approaches have been introduced to adaptively adjust the distribution

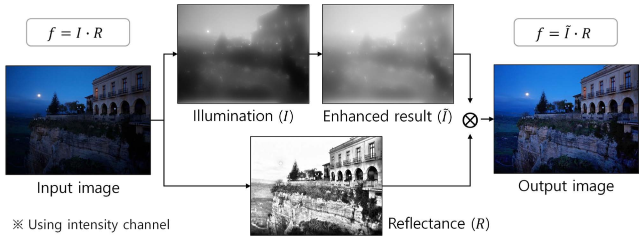

FIGURE 3. **Overall procedure of the decomposition model. Note that the enhanced result in this example is obtained by [21].**

of pixel values through histogram modifications and specifications [11], [12], [13]. As an example, Ueda and Suetake [13] transformed combination coefficients of the color vector space by utilizing the histogram specification technique while preserving color attributes of the original image. In order to involve the spatial information into the histogram, several approaches defined 2D histograms based on the relationship between neighbor pixels. Specifically, Celik and Tjahjadi [14] proposed to construct the 2D histogram using the mutual relationship between each pixel and its neighboring ones and conducted mapping the diagonal elements of the input histogram to those of the target histogram for image enhancement. Lee *et al.* [15] also computed the 2D histogram based on the relationship between neighbor pixels in a local region and utilized the layered difference for contrast enhancement. On the other hand, visually attentive regions have been adopted to efficiently suppress unexpected artifacts and noise when the histogram modification is conducted for low-light image enhancement [16].

a: IMAGE QUALITY-COMBINED HISTOGRAM METHODS

Since the coherence with the perceptual quality is one of important factors to evaluate the performance of low-light image enhancement, several studies have attempted to apply the image quality assessment (IQA) to the enhancement process. Most representatively, Gu *et al.* [66] proposed to combine subjective and objective assessments to derive the optimal histogram mapping based on both the entropy from the phase congruency and the momentum from pixel statistics. They further exploited the quality optimization scheme, which guides histogram modification to successively rectify image brightness and contrast to the proper level [18]. It is noteworthy that such attempts are highly suggestive for the learning-based approaches in that they reflect subjective evaluations during the enhancement process without severely enforcing the bias in terms of the limited number of experts to the learning procedure.

Even though histogram-based methods are conceptually simple and easy to implement, the downside lies in lack of the spatial correlation, that is, the position information of pixels is still hardly reflected to the enhancement process.

2) DECOMPOSITION MODELS

Based on the principle that the image f can be decomposed into two components, i.e., illumination (I) and reflectance (R), many researchers have been thinking about how to separate the illumination component from a single image. If this task is successfully done, the estimated lighting condition can be adjusted independently of the scene reflectance without any significant distortion. However, this problem is ill-posed (i.e., a single given input for two unknown parameters) as mentioned, thus the prior information on the illumination component has been strongly required. The overall procedure of the decomposition model is shown in Fig. 3.

a: OPTIMIZATION-BASED METHODS

The baseline of the optimization-based method can be formulated as follows:

$$\arg\min_{I,R} ||I \cdot R - f||_F^2 + \lambda ||\nabla I||_1, \tag{2}$$

where λ denotes the balancing factor. $||\cdot||_F$ and $||\cdot||_1$ represent the Frobenius norm and L_1 norm, respectively. Note that other regularization terms can be additionally used (e.g., the gradient of the reflectance). To resolve this under-constrained problem smoothly, various optimization techniques have been introduced. Specifically, Kimmel *et al.* [19] proposed to adopt the variational framework with the gradient of the illumination component (i.e., total variation [20]) as shown in the second term of (2). This scheme has shown the reliable decomposition results, however, noisy elements under dark areas are often amplified particularly in highly textured regions due to the derivative process in the log domain. To suppress such undesirable artifacts, Fu *et al.* [21] proposed a weighted variational framework by imposing the exponential factors to gradients of illumination and reflectance components, which efficiently alleviate the logarithmic effect of the derivative operation. By doing this, their approach is able to estimate illumination and reflectance simultaneously as shown in Fig. 3. Guo *et al.* [22] also designed the structure prior to preserve the overall outline while smoothing out textural details. It is noteworthy that they further implement their optimization process in the frequency domain for speed-up of the proposed model. In addition to this, Li *et al.* [23] devised a novel regularizer to achieve structure-revealing low-light image enhancement while preserving the boundary of objects. More recently, Ren *et al.* [24] injected the low-rank prior into the Retinex decomposition process to efficiently avoid the problem of noise amplification in the reflectance map, which often occurs when calculating illumination and reflectance simultaneously. To this end, they designed the sequential optimization scheme, i.e., reflectance estimation after illumination estimation. Similarly, Hao *et al.* [25] aim at suppressing noise in the reflectance map by exploiting the semi-decoupled Retinex decomposition technique. They applied the Gaussian total variation to estimate the illumination map in an edge-preserving way while the reflectance is also gradually extracted with the original input and the intermediate result of illumination estimation. Those optimization-based approaches are likely to yield different

FIGURE 4. Overall framework for learning-based approaches. Left: reference-based model. Right: no-reference model. Note that reference-based models require paired data, which is challenging to simultaneously acquire low-light and normal images in practice, whereas no- reference ones perform without such constraints.

enhancement results according to the philosophy of designing the prior for illumination or reflectance.

b: IMAGE DESCRIPTOR-BASED METHODS

As a pioneer, Jobson *et al.* [26] proposed to estimate the reflectance by computing the difference between input and its Gaussian-smoothed version in the log space, and used the corresponding result as the illumination-invariant enhancement result, so-called *Retinex*. They further extend it to the multiscale framework to consider various bandwidths of Gaussians [27]. In order to pay more attentions on the naturalness of the enhancement result, image descriptors have been developed based on various motivations and observations. Specifically, Wang *et al.* [28] proposed a bright-pass filter for the decomposition task and conducted bi-log transformation to make a balance between textural details and naturalness. They also utilized the maximum value among RGB channels at each pixel position as the initial illumination component and this strategy has been popularly employed in other approaches. Liang *et al.* [29] applied the nonlinear diffusion technique to estimate the illumination component and carried out Gamma correction with histogram clipping for refining the corresponding estimated result. This approach is effective to suppress unnecessary noise via the conductance function during the diffusion process. Kim *et al.* [30] also proposed to adopt the nonlinear diffusion scheme and define the illumination map by selecting the maximum value at each pixel position during the diffusion process. This method is extended to the concept of the diffusion pyramid for taking into account unknown scales of various objects more accurately [31]. On the other hand, there have been many attempts to explore illumination-invariant features for advanced applications in the field of computer vision, e.g., background subtraction [32], [33], face recognition [34], [35], etc. Among them, the orthogonal transform, e.g., discrete cosine transform (DCT), singular value decomposition (SVD), etc., has shown the possibility for extracting the underlying structure regardless of lighting conditions. Inspired by such decomposition ability, the SVD-based principal component of the intensity lattice defined in a small local region also has been adopted for illumination estimation [36]. Even though most image descriptor-based methods have the analytic solution for estimating the illumination map, those are highly dependent on specific conditions of the given image, thus still have respective limitations.

B. LEARNING-BASED APPROACHES

1) REFERENCE-BASED MODEL

As seen above, the problem of low-light image enhancement is still difficult to solve even with sophisticated optimization frameworks and fine-grained image descriptors. Inspired by the great success of deep learning for the task of image recognition, several researchers started to adopt the deep neural network (DNN) to infer the relationship between low-light inputs and normal outputs (see Fig. 4). That is, the problem of low-light enhancement is narrowed down the regression problem. For training parameters of the network, the simple way is to use image pairs of low-light input and the target output, i.e., reference, with appropriate loss functions. Most learning-based approaches have been introduced via this reference-based strategy. Specifically, Ignatov *et al.* [37] simply stacked convolutional layers to generate the enhancement result while considering the quality of color, texture, and content based on corresponding loss functions. In [38], authors extracted features by utilizing stacked convolution layers and simultaneously conducted enhancement with each output of convolution layers based on the conventional encoder-decoder architecture. The final output is generated by fusing such intermediate enhancement results.

Beyond the simple stacking approach, various DNN architectures have been developed to restore the enhanced version of the low-light input more accurately. As an example, the adversarial learning helps the output be more realistic. Yang *et al.* [39] applied the perceptual quality-driven adversarial learning scheme to further improve the intermediate result with recursive band representations. As another example, Wang *et al.* [40] designed the backprojection module, which iteratively conduct lightening and darkening processes to learn the residual for normal-light estimations. On the other hand, decomposition-based learning models also have been studied. Cai *et al.* [41] first decomposed the given image into luminance and detail components by using the weighted least square scheme and each result is fed into two different convolution neural networks (i.e., U-net [42] for luminance and ResBlock [43] for detail) for improving the

TABLE 2. Summary of representative methods both in handcrafted feature-based and learning-based approaches. Note that source codes for many other methods are also released in the public domain.

Method	Paper title	Category	Date	Publication	Code
LDR [15]	Contrast enhancement based on layered difference representation of 2d histograms	Handcrafted	2013	TIP	Matlab
NPE [28]	Naturalness preserved enhancement algorithm for non-uniform illumination images	Handcrafted	2013	TIP	Matlab
SRIE [21]	A weighted variational model for simultaneous reflectance and illumination estimation	Handcrafted	2016	CVPR	Matlab
LIME [22]	LIME: low-light image enhancement via illumination map estimation	Handcrafted	2017	TIP	Matlab
HDRNet [48]	Deep bilateral learning for real-time image enhancement	Learning	2017	TOG	TensorFlow
DnR [54]	Distort-and-recover: color enhancement using deep reinforcement learning	Learning	2018	CVPR	TensorFlow
DPE [79]	Deep photo enhancer: unpaired learning for image enhancement from photographs with GANs	Learning	2018	CVPR	TensorFlow
DeepUPE [44]	Underexposed photo enhancement using deep illumination estimation	Learning	2019	CVPR	TensorFlow
Zero-DCE [2]	Zero-reference deep curve estimation for low-light image enhancement	Learning	2020	CVPR	PyTorch
DeepLPF [49]	DeepLPF: deep local parametric filters for image enhancement	Learning	2020	CVPR	PyTorch
DRBN [39]	From fidelity to perceptual quality: a semi-supervised approach for low-light image enhancement	Learning	2020	CVPR	PyTorch
DSLR [51]	DSLR: deep stacked Laplacian restorer for low-light image enhancement	Learning	2021	TMM	PyTorch
EnlightenGAN [58]	EnlightenGAN: deep light enhancement without paired supervision	Learning	2021	TIP	PyTorch

corresponding results, respectively. Wang *et al.* [44] devised the encoder-decoder architecture to infer the illumination component in an implicit way instead of directly generating the enhancement result. Zhang *et al.* [45] exploited the decomposition layer with the two-branch architecture. Specifically, the estimated reflectance is restored through the traditional encoder-decoder stream while the illumination is adjusted with stacked convolution layers. Xu *et al.* [46] proposed a frequency-based decomposition scheme, that is, their encoder-decoder architecture first learns to recover image objects in the low-frequency layer and subsequently enhance high-frequency details based on the recovered object image. Yang *et al.* [47] formulated the decomposition problem as the problem of sparse gradient minimization on the initially extracted illumination map, which can be solved by using the Prewitt gradient operator in a fully learnable way via their network architecture. In another research direction, several studies have attempted to combine multiscale features for considering global and local properties separately during the enhancement process. Gharbi *et al.* [48] applied the concept of bilateral filtering to guide the enhancement process in the low-resolution space. In turn, the high-resolution stream learns affine coefficients back to the enhanced output. Moran *et al.* [49] proposed to utilize learned filters of three different types, i.e., polynomial, graduated, and elliptical filters, for global and local enhancement operations separately. Li *et al.* [50] designed a luminance-aware pyramid network in a coarse-to-fine manner, which is composed of two coarse feature extractors and the refinement branch in the fine scale. Lim and Kim [51] proposed to restore the global layout and local details separately based on the Laplacian pyramid both in image and feature spaces. In [52], authors also proposed a multi-branch topology residual block to grasp global and local features more efficiently.

Even though learning-based approaches have shown remarkable improvements for low-light image enhancement, there is a clear limitation in that enhancement results are strongly affected by the characteristic of the training data, which is constructed based on the subjective re-touch process or the artificial adjustment of the exposure time. To cope with the learning bias, several attempts have been made without using paired low/normal images, which will be explained in detail in the following subsection.

2) NO-REFERENCE MODEL

In this category, researchers have devoted considerable efforts to achieve the generality regardless of training datasets. To do this, most approaches do not use pairs of low-light input and re-touched normal output. Specifically, Chen *et al.* [79] designed the two-way generative adversarial network to reflect the naturalness of the reconstruction result by unpaired learning as well as the consistency with the underlying properties of the original image in a bi-directional manner. Park *et al.* [54] designed the deep reinforcement learning scheme based on the Markov decision process. In particular, their agent network conducted the distort-and-recover process only with high-quality normal images (i.e., without using the paired dataset) during the training phase. Kosugi and Yamasaki [55] adopted the generative adversarial network with the reinforcement learning framework where the generator works as the agent that selects parameters of Adobe Photoshop and is rewarded when it fools the discriminator. That is, by utilizing the Wassersatein distance [56], the enhancement result is compared with different high-quality images to compute the reward score. Similarly, Ni *et al.* [57] proposed to inject the quality attention module into the generator such that it learns domain-relevant quality attention directly from spatial and style characteristics with unpaired data. Jiang *et al.* [58] proposed to regularize the unpaired training scheme by using information extracted from the input itself, which is based on the global-local discriminator with the self-regularized perceptual loss and the attention mechanism. On the other hand, Guo *et al.* [2] attempted to estimate the mapping curve for the global adjustment without any paired or unpaired data in a pixel-wise manner. To do this, a set of no-reference loss functions are carefully designed, which implicitly evaluate the visual quality of the enhancement result. They further present a light-weight version of their previous work that takes advantage of the tiny network with just 10K parameters, which can be applied to the embedded platforms in real-world applications [59]. Zhu *et al.* [60] proposed a zero-shot scheme based on three-branch convolutional neural network, which is optimized by iteratively minimizing a specially designed loss function without any prior of image samples.

Even though such no-reference models are appealing in its relaxed assumption on reference images, the enhancement performance is somewhat inferior to the reference-based methods. To bridge the gap between the performance and the generality, some researchers have introduced the semi-supervised approaches, i.e., paired and unpaired data are used together in a convenient way [39], [61], [62]. Due to the power of deep learning, many other methods still have been actively proposed based on deep neural architectures in literature.

C. SUMMARY

Until now, we have reviewed a variety of studies for low-light image enhancement in a balanced way between handcrafted feature-based and learning-based approaches. From this analysis, there are some remarks as follows:

- Most methods have consistently sought the optimal solution to resolve the ill-posedness nature of low-light image enhancement. Starting from the Retinex theory, various deep learning techniques have now been widely adopted to clarify this ambiguity.
- Even though learning-based approaches have brought huge performance gains, the learning bias has emerged as a new problem, i.e., the enhancement result is strongly affected by the characteristic of the dataset, which is used for training parameters of the network. To cope with this limitation, unsupervised learning schemes with zero-shot or unpaired data have been explored most recently. Ultimately, in order to guarantee the generality under diverse real-world scenarios, it is expected that the research will proceed in the direction of developing reliable algorithms that can be used without paired data.
- Although the majority in this field are adopting learning-based approaches, handcrafted feature-based methods are still required to effectively deal with various real-world situations with limited resources.

To improve the reproducibility and disseminate the research results efficiently, source codes of most methods for low-light image enhancement are available in the public domain. Table 2 shows the detailed information for several representative methods both in handcrafted feature-based and learning-based approaches.

IV. EXPERIMENTAL RESULTS

In this Section, we have demonstrated the performance of several representative methods introduced in the previous Section for better understanding about remaining challenges as well as effects of low-light image enhancement. Details of experimental results will be explained in the following subsections.

A. DATASETS

For the performance evaluation of low-light image enhancement, unpaired and paired datasets have been introduced in literature. The characteristics and statistical composition of each dataset can be summarized as follows:

1) NASA RETINEX

This dataset [63] is composed of 25 images whose resolution is 1312×2000 pixels (or 2000×1312 pixels). Samples contain various lighting conditions acquired both in indoor and outdoor environments, e.g., cast shadows, dim light, dark indoor, etc.

2) HDR

The HDR dataset [64] is constructed by a set of sequential exposures and the total seven images (900×1350 pixels or 1350×900 pixels) are provided for the test. Sample images contain a single person and are overall dark due to the low exposure.

3) NPE

This dataset [28] consists of eight images, which mainly contain the landscape scene captured under the cloudy condition. The resolution of samples varies from 267×304 to 749×492 pixels.

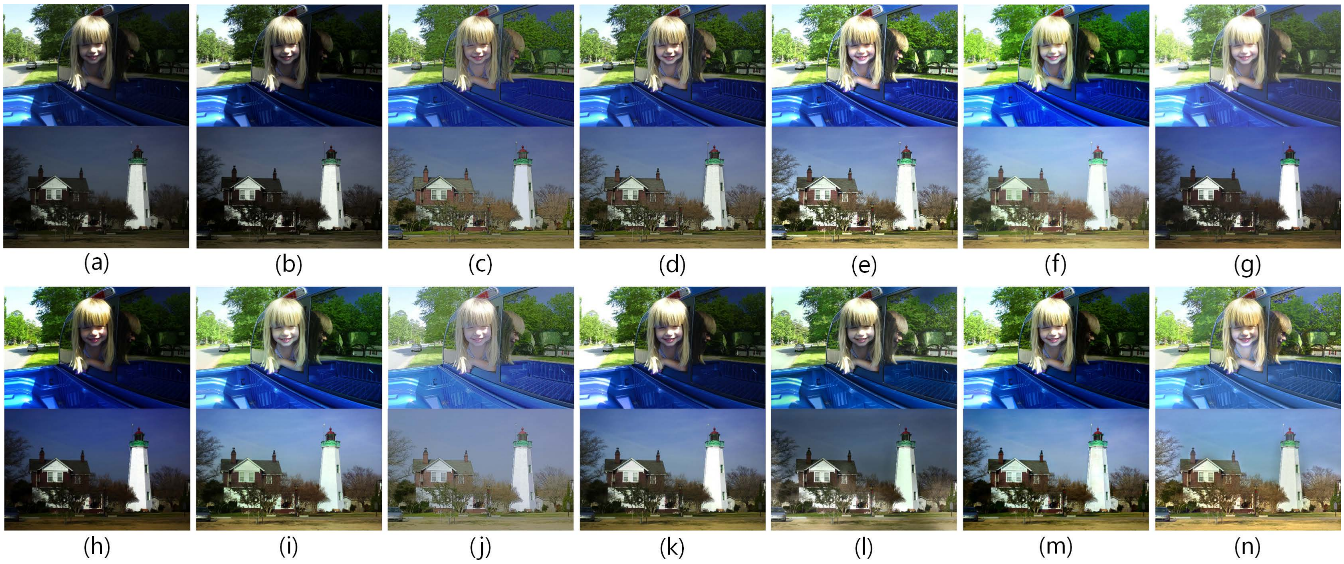

FIGURE 5. Several examples of low-light image enhancement on the NASA Retinex dataset [63]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58].

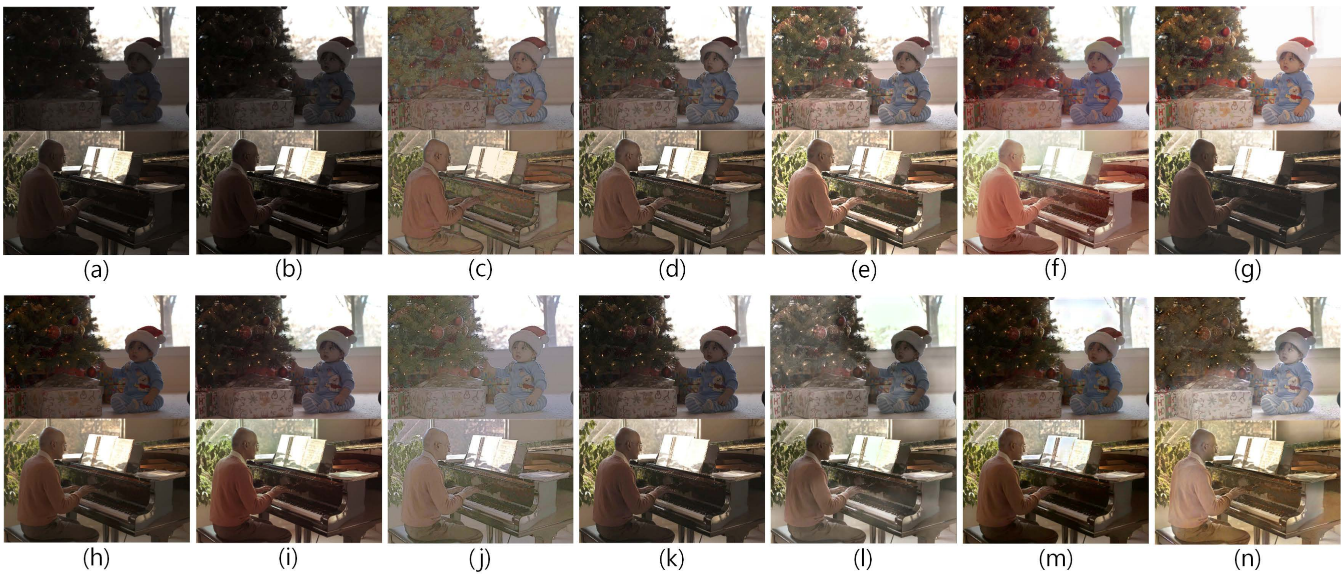

FIGURE 6. Several examples of low-light image enhancement on the HDR dataset [64]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58].

4) LIME

The LIME dataset [22] consists of 10 images whose resolution varies from 326 × 326 to 2000 × 1500 pixels. In particular, objects and background of various colors are mainly taken at the night time to check whether enhancement results preserve the color attribute as well as the vividness.

5) MIT-ADOBE FIVEK

This dataset [8] is one of the most popularly employed datasets for learning-based approaches. It consists of total 5000 pairs of under-exposed (i.e., low-light) images and corresponding enhanced results, which are re-touched by five experts. In general, 4500 pairs are used for training while the remaining 500 pairs are employed for test. The resolution of samples varies from 1440 × 2160 to 6048 × 4032 pixels.

6) SICE

The SICE dataset [41] contains 4413 multi-exposure images and the corresponding enhanced results are generated by combining results of image fusion algorithms. Note that subjective evaluations are subsequently conducted to screen the best quality one as the reference image of each scene.

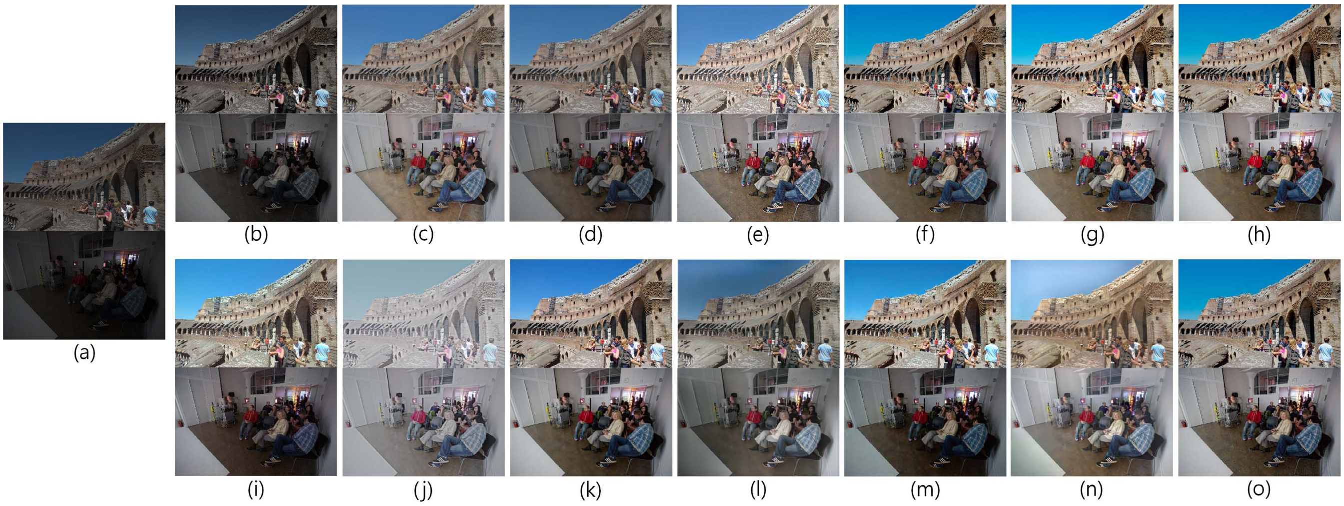

FIGURE 7. Several examples of low-light image enhancement on the MIT-Adobe FiveK dataset [8]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58]. (o) Ground truth.

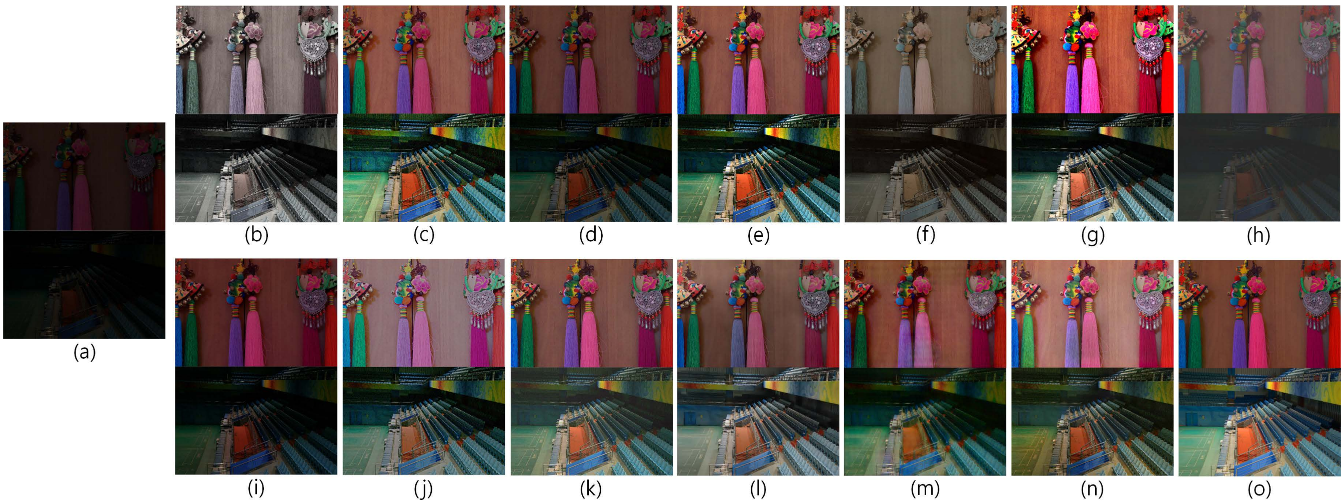

FIGURE 8. Several examples of low-light image enhancement on the LOL dataset [9]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58]. (o) Ground truth.

The resolution of most images are between 3000×2000 pixels and 6000×4000 pixels.

7) LOL

The LOL dataset [9] contains total 500 low/normal-light image pairs, which are acquired from real-world environments. For training, total 485 pairs are utilized while remaining 15 pairs are adopted for the validation. The resolution of all the samples is 400×600 pixels.

Note that NASA Retinex, HDR, NPE, and LIME do not provide the ground truth for input low-light images whereas MIT-Adobe FiveK, SICE, and LOL are paired datasets. In what follows, representative algorithms for low-light image enhancement will be tested based on four selected datasets among them, i.e., NASA Retinex, HDR, MIT-Adobe FiveK, and LOL datasets.

B. PERFORMANCE EVALUATION

Total 13 representative methods, i.e., LDR [15], NPE [28], SRIE [21], LIME [22], HDRNet [48], DnR [54], DPE [79], DeepUPE [44], Zero-DCE [2], DeepLPF [49], DRBN [39], DSLR [51], and EnlightenGAN [58] (four handcrafted feature-based approaches and nine learning-based approaches), for low-light image enhancement have been evaluated on selected benchmark datasets and detailed results are analyzed and discussed in this subsection.

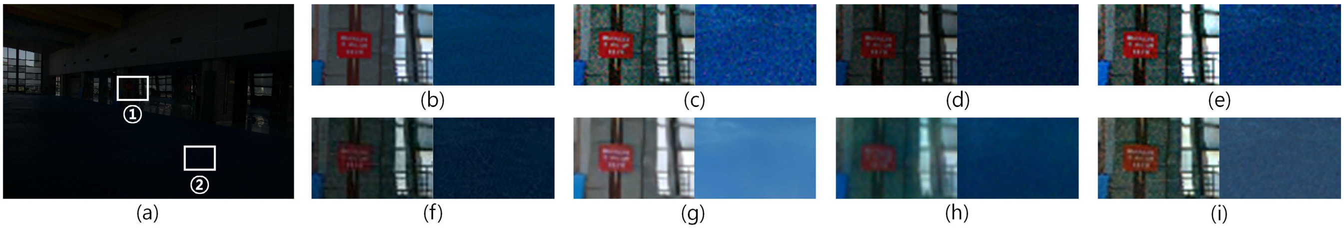

FIGURE 9. Example of the performance comparison in local regions. (a) Input selected from the LOL dataset [9]. Enlarged regions of by (b) Ground truth, (c) NPE [28], (d) SRIE [21], (e) LIME [22], (f) DeepLPF [49], (g) DRBN [39], (h) DSLR [51], and (i) EnlightenGAN [58]. Note that left and right sub-images are enhancement results for ① and ②, respectively.

TABLE 3. Performance comparison on NASA Retinex [63] and HDR [64] datasets. Note that the best result is in red while the second and the third results are in blue and green, respectively.

Method	NASA Retinex dataset			HDR dataset		
	NIQE(↓)	BTMQI(↓)	NIQMC(↑)	NIQE(↓)	BTMQI(↓)	NIQMC(↑)
Input	5.106±1.556	4.763±1.488	4.625±0.947	4.653±1.012	5.943±0.671	4.685±0.759
LDR [15]	4.458±1.059	4.527±1.422	5.129±0.483	4.326±0.920	5.541±0.851	4.952±0.634
NPE [28]	3.755±0.808	3.466±1.404	4.709±0.443	3.536±0.573	4.106±1.125	4.370±0.497
SRIE [21]	4.108±1.323	3.508±1.639	4.809±0.933	3.958±0.695	3.197±1.146	4.989±0.351
LIME [22]	4.018±1.201	4.549±1.383	5.097±0.864	3.582±0.632	4.618±0.964	5.027±0.176
HDRNet [48]	4.162±1.147	4.396±1.519	5.131±0.865	3.572±0.875	4.787±0.962	5.057±0.346
DnR [54]	4.421±1.124	4.534±1.693	4.899±0.854	3.975±0.814	5.536±0.962	4.917±0.265
DPE [79]	4.759±1.043	4.561±1.470	4.921±0.677	3.888±0.872	4.731±0.789	4.963±0.589
DeepUPE [44]	4.301±1.179	3.972±1.474	5.197±0.827	3.686±0.947	4.081±1.239	5.173±0.347
Zero-DCE [2]	4.275±1.268	4.231±1.765	4.484±1.112	4.326±0.752	5.001±1.115	4.445±0.347
DeepLPF [49]	4.673±1.198	3.974±1.571	5.084±0.936	4.458±0.965	4.909±1.159	4.866±0.608
DRBN [39]	4.768±1.333	3.373±1.319	5.041±0.765	4.261±0.877	3.059±0.767	5.189±0.202
DSLR [51]	3.858±0.839	3.922±1.413	5.279±0.760	3.249±0.685	3.819±0.964	5.274±0.272
EnlightenGAN [58]	3.920±1.441	4.516±1.323	4.829±1.239	3.593±0.579	4.228±1.221	5.048±0.163

1) QUALITATIVE EVALUATION

First of all, enhancement results by 13 methods on NASA Retinex and HDR datasets are shown in Figs. 5 and 6, respectively. Since these datasets do not provide training samples, learning-based methods are trained by using the MIT-Adobe FiveK dataset while other parameter settings are not modified. Specifically, the effect of casting shadows (top) and cloudy weather (bottom) shown in Fig. 5 is successfully mitigated in enhancement results by most methods. However, LDR and DPE yield somewhat conservative enhancement results while NPE, Zero-DCE, DRBN, and EnlightenGAN show wash-out artifacts, which make enhancement results unrealistic. For very dark indoor images, LIME, HDRNet, and DnR provide enhancement results more vividly compared to other methods as shown in Fig. 6. DeepUPE, DeepLPF, and DSLR also show visually satisfactory results without any severe distortion in background. It is noteworthy that enhancement results of learning-based approaches are strongly dependent on the property of training samples. Thus, their results, which are generated from the model trained with the MIT-Adobe FiveK dataset in this case, are probably vulnerable to unseen samples.

In the following, the performance of previous methods is evaluated on paired datasets, i.e., MIT-Adobe FiveK and LOL datasets. Basically, learning-based approaches show better performance compared to the test results for NASA Retinex and HDR datasets since those are trained with samples of MIT-Adobe FiveK and LOL datasets. The corresponding results are shown in Figs. 7 and 8, respectively. Specifically, LDR, NPE, and SRIE provide under-enhanced results compared to the ground truth whereas LIME is likely to exaggerate local details as shown in Fig. 7(e). From these results, we can see that handcrafted feature-based approaches often have difficulties to accurately restore color attributes (see the blue sky in the first example (top) of Fig. 7) of the given scene. In general, learning-based approaches show reliable enhancement results while successfully keeping the characteristics of the original scene. Note that Zero-DCE and DRBN still suffer from wash-out artifacts as shown in Fig. 7(j) and (l). This tells us that unsupervised (or semi-supervised) schemes still have challenges to overcome such limitation. Figure 8 shows enhancement results on the LOL dataset, which contains extremely dark images as input. In particular, DeepLPF shows the almost same result with the ground truth (see Fig. 8(k)) whereas HDRNet fails to recover the color attribute of the original scene. Among handcrafted feature-based approaches, LIME provides the reliable results especially under the very dark environment while keeping the color vividness (see Fig. 8(e)). Moreover, local enhancement results are also presented in Fig. 9. As can be seen, under very dark lighting condition, several methods suffer from noise amplifications (see Fig. 9(c), (e), and (i)). In this example,

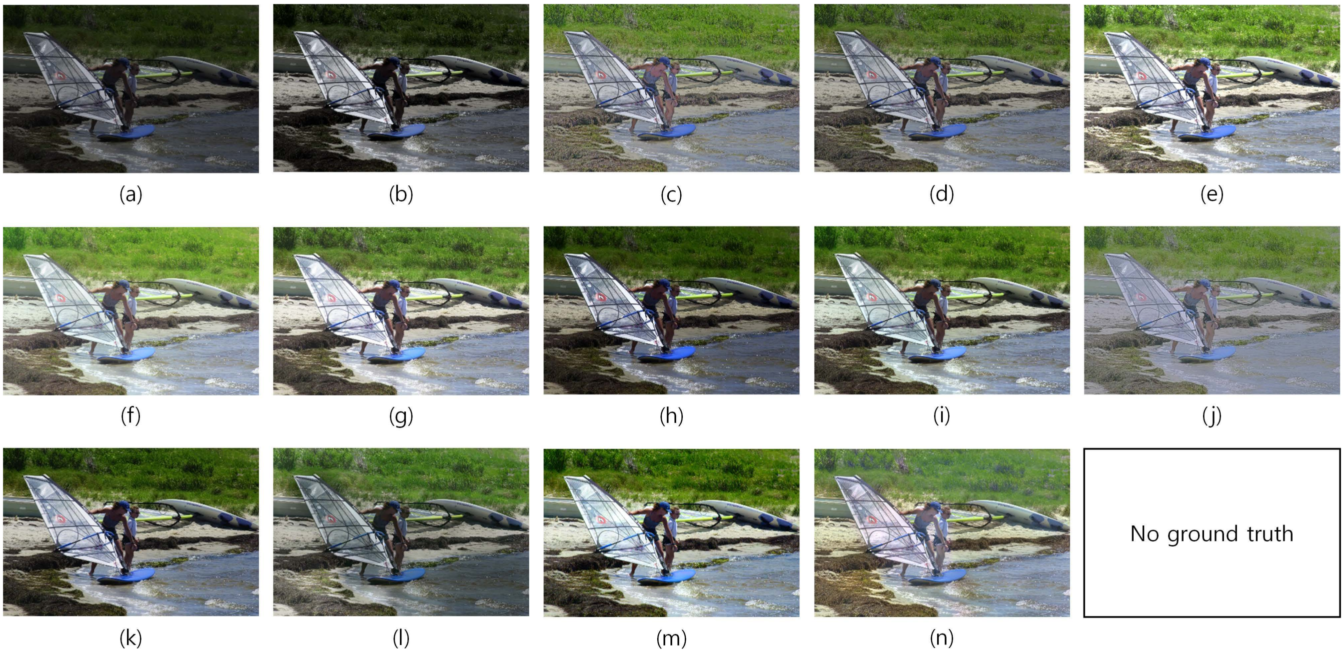

FIGURE 10. More enhancement results on the NASA Retinex dataset [63]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58].

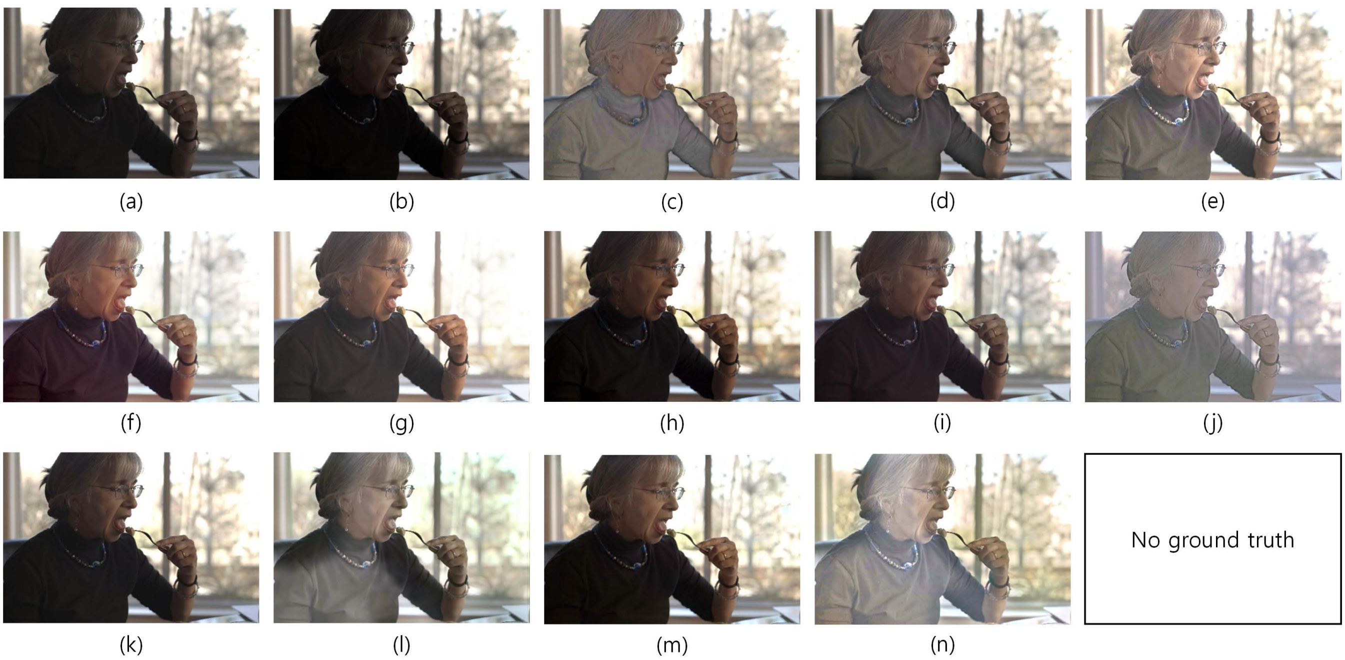

FIGURE 11. More enhancement results on the HDR dataset [64]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58].

DRBN fails to recover the color attribute of the original scene although it efficiently suppresses undesirable noise. The qualitative performance of each method can be summarized as follows:

- **LDR**: It mostly generates under-enhanced results since this model is designed to consider the global adjustment without severe distortions in dark background.
- **NPE**: NPE efficiently suppresses low-lighting effects in enhancement results whereas this method often fails to restore the color properties of the underlying scene.
- **SRIE**: It produces reasonable results in most cases without significant variations from image to image since SRIE attempts to simultaneously estimate illumination

FIGURE 12. More enhancement results on the MIT-Adobe FiveK dataset [8]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58]. (o) Ground truth.

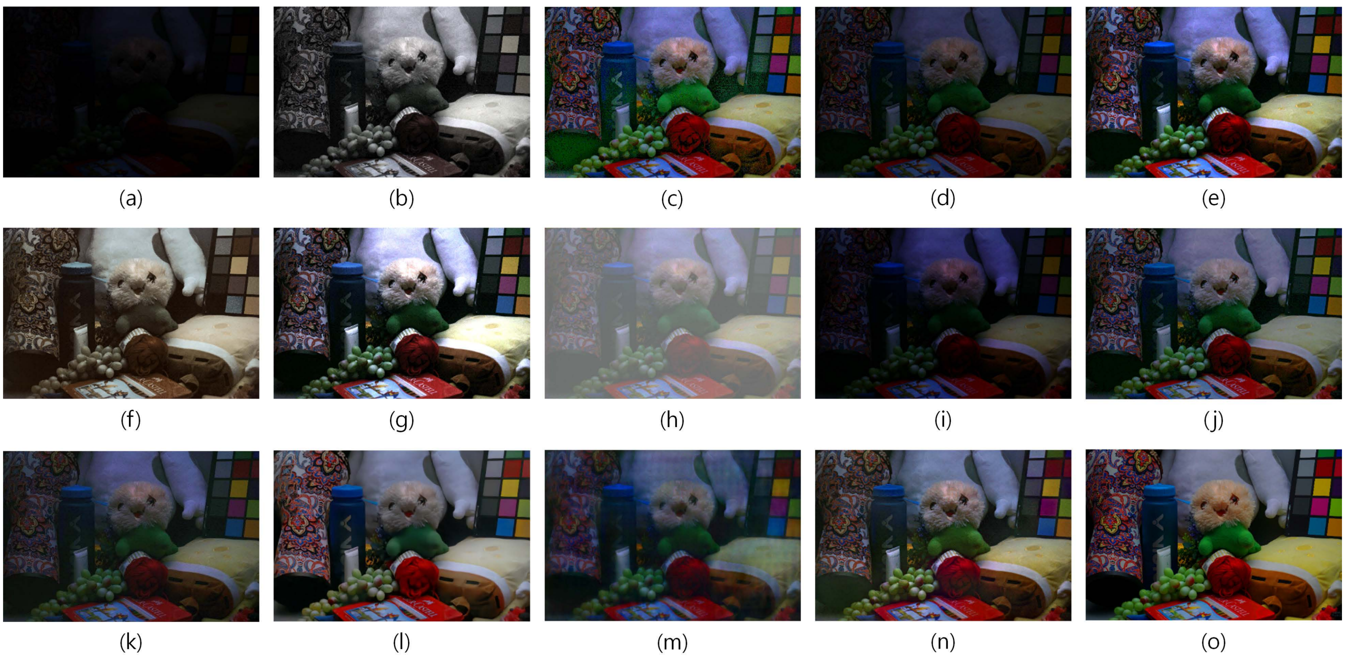

FIGURE 13. More enhancement results on the LOL dataset [9]. (a) Input. (b) LDR [15]. (c) NPE [28]. (d) SRIE [21]. (e) LIME [22]. (f) HDRNet [48]. (g) DnR [54]. (h) DPE [79]. (i) DeepUPE [44]. (j) Zero-DCE [2]. (k) DeepLPF [49]. (l) DRBN [39]. (m) DSLR [51]. (n) EnlightenGAN [58]. (o) Ground truth.

and reflectance. However, it often fails to restore very dark images.

- **LIME**: This method tends to exaggerate the contrast of enhancement results, however, the color vividness is successfully restored.
- **HDRNet**: It aims to adjust the dynamic range of low-light images for visual enhancement, thus HDRNet generates moderate enhancement results in most cases. For extremely low-lighting environments, it often fails to restore color attributes.

TABLE 4. Performance comparison on the MIT-Adobe FiveK [8] dataset. The performance comparison is conducted with both full-reference and no-reference VQMs. Note that the best result is in red while the second and the third results are in blue and green, respectively.

Method	PSNR(↑)	SSIM(↑)	NIQE(↓)	BTMQI(↓)	NIQMC(↑)
Input	-	-	5.618±1.483	6.278±0.759	3.493±0.703
LDR [15]	18.61±4.205	0.734±0.107	3.676±1.139	4.765±1.188	4.850±0.476
NPE [28]	17.28±3.251	0.807±0.108	3.929±1.006	3.952±1.157	4.628±0.348
SRIE [21]	18.65±2.395	0.837±0.072	3.930±1.134	4.664±1.388	4.244±0.591
LIME [22]	16.07±2.579	0.807±0.101	3.359±1.058	4.475±1.197	4.787±0.583
HDRNet [48]	22.04±7.063	0.855±0.072	4.329±1.194	4.606±1.199	5.098±0.366
DnR [54]	20.89±4.732	0.815±0.053	4.480±1.218	4.493±1.351	5.174±0.236
DPE [79]	22.21±5.582	0.852±0.073	4.229±1.394	4.367±1.132	5.149±0.317
DeepUPE [44]	23.10±3.885	0.878±0.057	4.224±1.213	4.301±1.341	5.212±0.523
Zero-DCE [2]	12.82±2.073	0.721±0.101	3.845±1.049	5.529±1.155	4.093±0.751
DeepLPF [49]	23.92±4.404	0.884±0.003	4.177±2.143	4.679±1.260	4.717±0.429
DRBN [39]	16.37±2.628	0.762±0.093	4.382±1.380	3.941±1.247	4.751±0.530
DSLR [51]	24.34±3.236	0.904±0.035	4.209±1.615	4.298±1.048	5.209±0.306
EnlightenGAN [58]	15.58±2.777	0.759±0.095	3.767±1.043	4.324±1.269	4.766±0.525

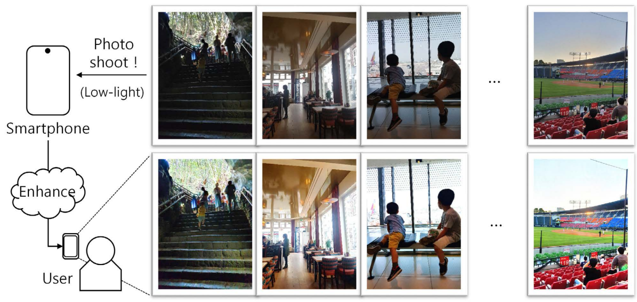

FIGURE 14. Example of the smartphone application. Note that original pictures are taken with Samsung Galaxy S10+ (top) and corresponding enhancement results are generated by DSLR [51] (bottom).

- **DnR**: DnR produces the high contrast results while restoring the color vividness for all the datasets. Even though DnR shows reasonable enhancement results, it makes normal-lighted areas in the original image be over-saturated.
- **DPE**: This model generates moderate enhancement results in most cases without any significant distortion, however, the enhancement results are somewhat under-enhanced for very dark images.
- **DeepUPE**: DeepUPE works qualitatively similar to DnR and DPE, thus it also shows visually-pleasing results for various samples except for the very dark images.
- **Zero-DCE**: Even though Zero-DCE works regardless of the learning bias, it still shows the performance gap compared to supervised approaches. Specifically, Zero-DCE is likely to yield wash-out results in most cases.
- **DeepLPF/DRBN/DSLR**: Those models generally produces visually acceptable enhancement results in most cases even under extremely low-lighting conditions.
- **EnlightenGAN**: Similar to Zero-DCE, it often shows wash-out results. However, it works quite reliably for very dark environments.

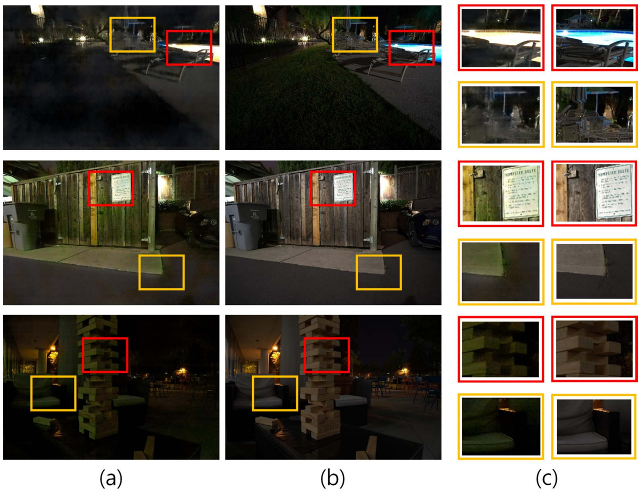

FIGURE 15. Several examples of enhancement results from extremely low-light images (SID dataset [79]). (a) RED method [80]. (b) Ground truth. (c) Enlarged regions selected from (left: (a)) and (right: (b)), respectively. Note that color attributes are somewhat distorted in restored images.

Based on qualitative evaluations, it is naturally thought that the performance of learning-based approaches is strongly affected by properties of training samples. That is, the performance of low-light image enhancement is quite satisfactory in the intra-dataset test, however, images taken from real-world scenarios are mostly unseen during training, which probably leads to the performance drop. Moreover, unsupervised learning-based approaches show somewhat unstable results according to the degree of darkness. On the other hand, several handcrafted feature-based approaches are still able to restore visually-pleasing enhancement results without any learning process. Even though the detailed characteristics of the original scene, e.g., color attributes and vividness, light intensity, etc., are often incorrectly restored, such methods are capable of working consistently based on their

TABLE 5. Performance comparison on the LOL [9] dataset. The performance comparison is conducted with both full-reference and no-reference VQMs. Note that the best result is in red while the second and the third results are in blue and green, respectively.

Method	PSNR(↑)	SSIM(↑)	NIQE(↓)	BTMQI(↓)	NIQMC(↑)
Input	-	-	6.749±1.110	6.651±0.275	1.602±0.551
LDR [15]	15.89±2.616	0.548±0.093	8.336±1.104	7.875±1.224	4.654±0.289
NPE [28]	17.86±3.463	0.748±0.138	8.439±1.213	2.971±0.886	4.692±0.317
SRIE [21]	11.94±4.044	0.563±0.182	7.287±1.078	5.634±1.034	3.489±0.287
LIME [22]	17.71±4.521	0.740±0.162	8.129±1.173	3.013±0.702	5.077±0.199
HDRNet [48]	15.32±4.244	0.613±0.138	7.801±1.311	4.057±1.371	4.428±0.714
DnR [54]	16.03±2.768	0.635±0.167	8.137±1.136	3.563±1.208	5.217±0.611
DPE [79]	14.23±3.884	0.532±0.172	5.069±0.969	5.799±0.747	2.890±1.105
DeepUPE [44]	11.04±3.930	0.429±0.191	7.366±1.071	6.287±0.903	3.477±0.511
Zero-DCE [2]	17.09±4.844	0.565±0.123	7.929±1.109	3.379±1.069	4.416±0.223
DeepLPF [49]	14.93±5.288	0.646±0.177	3.667±0.496	4.714±1.079	4.077±0.486
DRBN [39]	16.78±3.821	0.701±0.049	5.109±0.965	3.145±0.741	4.771±0.246
DSLR [51]	15.24±5.545	0.624±0.198	5.334±0.602	4.644±0.697	4.262±0.233
EnlightenGAN [58]	17.88±5.075	0.726±0.171	4.684±0.775	2.813±0.883	4.682±0.232

design philosophies. More enhancement results on each benchmark are provided in Figs. 10, 11, 12, and 13.

2) QUANTITATIVE EVALUATION

To quantitatively evaluate the performance of low-light image enhancement, the visual quality metrics (VQMs) have been widely employed. Specifically, for the case of paired datasets, two representative full-reference metrics, i.e., PSNR and SSIM, have been used, which compute the difference between the enhancement result and the ground truth in a pixel-wise manner. For unpaired datasets, no-reference metrics, i.e., metrics based on computation of the quality score without the ground truth, have been adopted for the performance evaluation. Among many no-reference metrics, which have been developed in the field of image quality assessment (IQA), three representative metrics, i.e., NIQE [65], BTMQI [66], and NIQMC [67], are applied to measure the visual quality of the enhancement result in this review. Specifically, NIQE focuses on estimating the naturalness of the enhancement result by computing the perceptual score, which is driven from the natural scene statistics. BTMQI represents the accuracy of color tone mapping from reference (i.e., low-light image) to target (i.e., enhancement result). NIQMC utilizes the visual attention mechanism and computes the quality score based on such attentive regions, which is highly relevant to the human vision system. Note that other metrics, e.g., LOE [28], C-PCQI [18], LPIPS(Learned Perceptual Image Patch Similarity) [68], CaHDC [69], etc., also can be employed for the performance evaluation. Based on such VQMs, the performance of each method can be analyzed from various perspectives.

The quantitative evaluation on NASA Retinex and HDR datasets is shown in Table 3. Since such two datasets do not provide the ground truth, three no-reference metrics, i.e., NIQE, BTMQI, and NIQMC, are used for the performance comparison. On average, NPE, SRIE, DRBN, and DSLR demonstrate the reliable results for NASA Retinex and HDR datasets. For example, DSLR is included in the top-3 performance for the NASA Retinex dataset except for the BTMQI metric. This is because the color properties of training (i.e., MIT-Adobe FiveK) and test (i.e., NASA Retinex) datasets are quite a different each other while the difference of the textural statistics is negligible. Therefore, DSLR results show somewhat the dropped performance for the BTMQI metric in this case. Even though learning-based methods are trained with other samples, which do not belong to NASA Retinex or HDR datasets, several methods still show competitive results for low-light image enhancement. In the following, the performance comparison on MIT-Adove FiveK and LOL datasets is shown in Tables 4 and 5, respectively. As expected, learning-based methods present better performance compared to handcrafted feature-based approaches in terms of full-reference metrics, i.e., PSNR and SSIM, since samples selected from the intra-dataset are employed for training each network. In particular, DeepUPE, DeepLPF, and DSLR show the reliable performance in the MIT-Adobe FiveK dataset whereas several unsupervised learning-based models often fail to appropriately restore the scene nature. For the LOL dataset, some handcrafted feature-based approaches, e.g., NPE and LIME show the competitive results with learning-based methods as shown in Table 5. In particular, test samples of the LOL dataset are very dark (see Fig. 8), thus values of PSNR or SSIM metrics are relatively low compared to those for the MIT-Adobe FiveK dataset.

3) IMPLEMENTATION ENVIRONMENT

For implementation of handcrafted feature-based approaches introduced in this review, Matlab codes, which are provided by authors, are utilized on the high-end PC (Intel Xeon 2.2 GHz CPU and 64 GB RAM). To train and test learning-based methods, a single PC equipped with an Intel i7-6850K@3.6 GHz CPU and a single NVIDIA GeForce RTX 3090 GPU is used. Note that parameters for

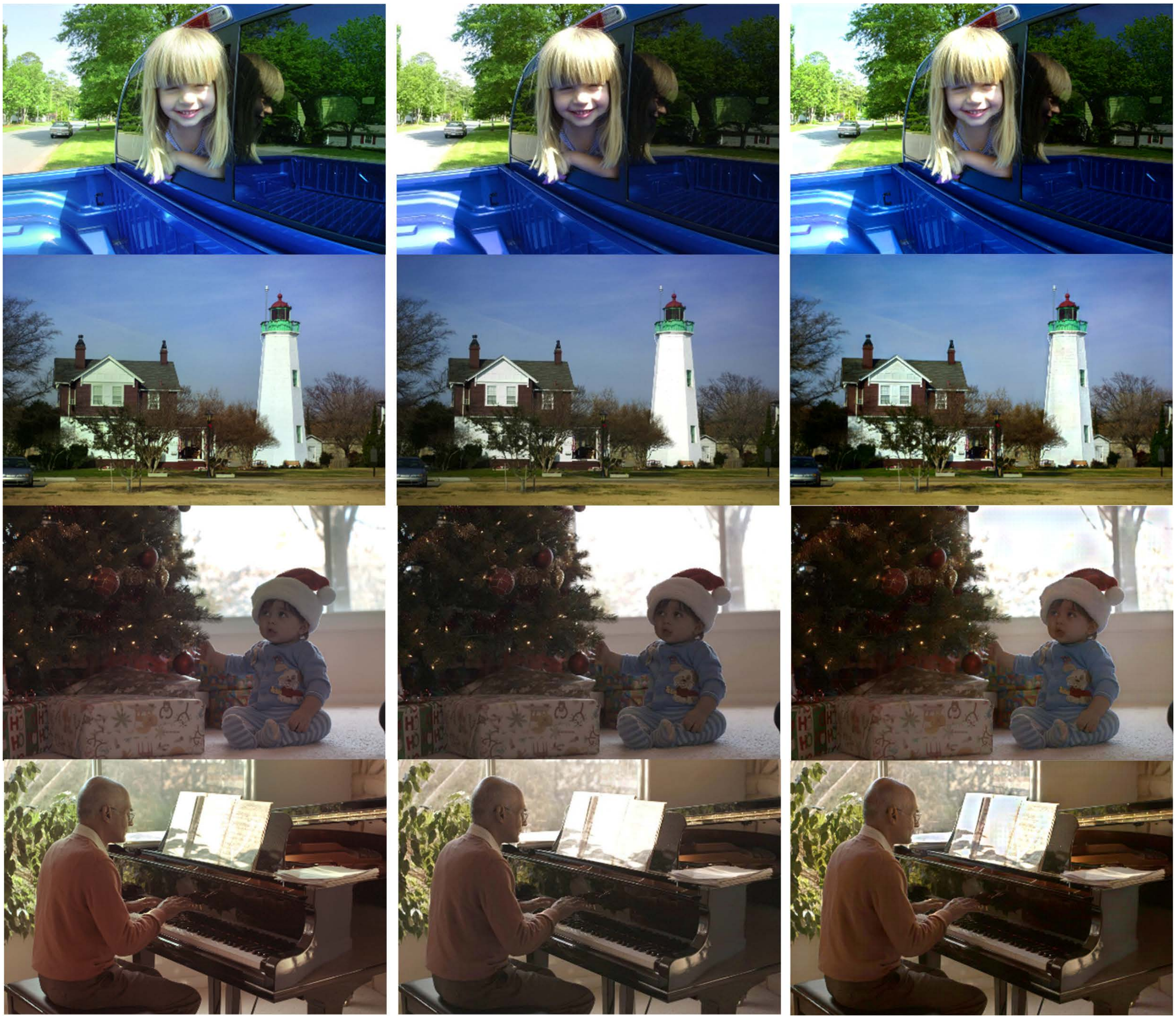

FIGURE 16. Enhance results by DeepUPE(left), DeepLPF(middle), and DSLR(right). Note that these results are enlarged versions from Figs. 5 and 6.

learning-based methods are not modified from source codes provided by authors.

C. DISCUSSION

Beyond simply stretching the dynamic range of the pixel value (e.g., Gamma correction, histogram equalization, etc.), realistic lighting restoration has been actively explored via the power of deep learning. This technique now becomes essential to improve the user experience for various embedded platforms as well as home-appliance services. For example, the visual quality of low-light images taken with smartphones can be efficiently improved without using additional hardware sensors as shown in Fig. 14. This also gives a great help to improve the performance of other vision tasks, e.g., object detection and recognition in the dark [70], [71], [72]. Even though the ill-posedness nature of low-light image enhancement has been alleviated by inferring the relationship between low-light/normal paired images based on the large amount of training samples, the learning bias, i.e., dependency on the training dataset, still makes the performance drop in diverse cases of real-world scenarios. To cope with existing limitations and construct the reliable enhancement system, sufficient discussions are required on following issues:

- **Algorithm.** In general, the majority of low-light image enhancement is based on the learning-based approaches. For generality of the enhancement performance, the research trend is beginning to shift towards unsupervised learning approaches. Several studies already have concentrated on developing the learning strategy with

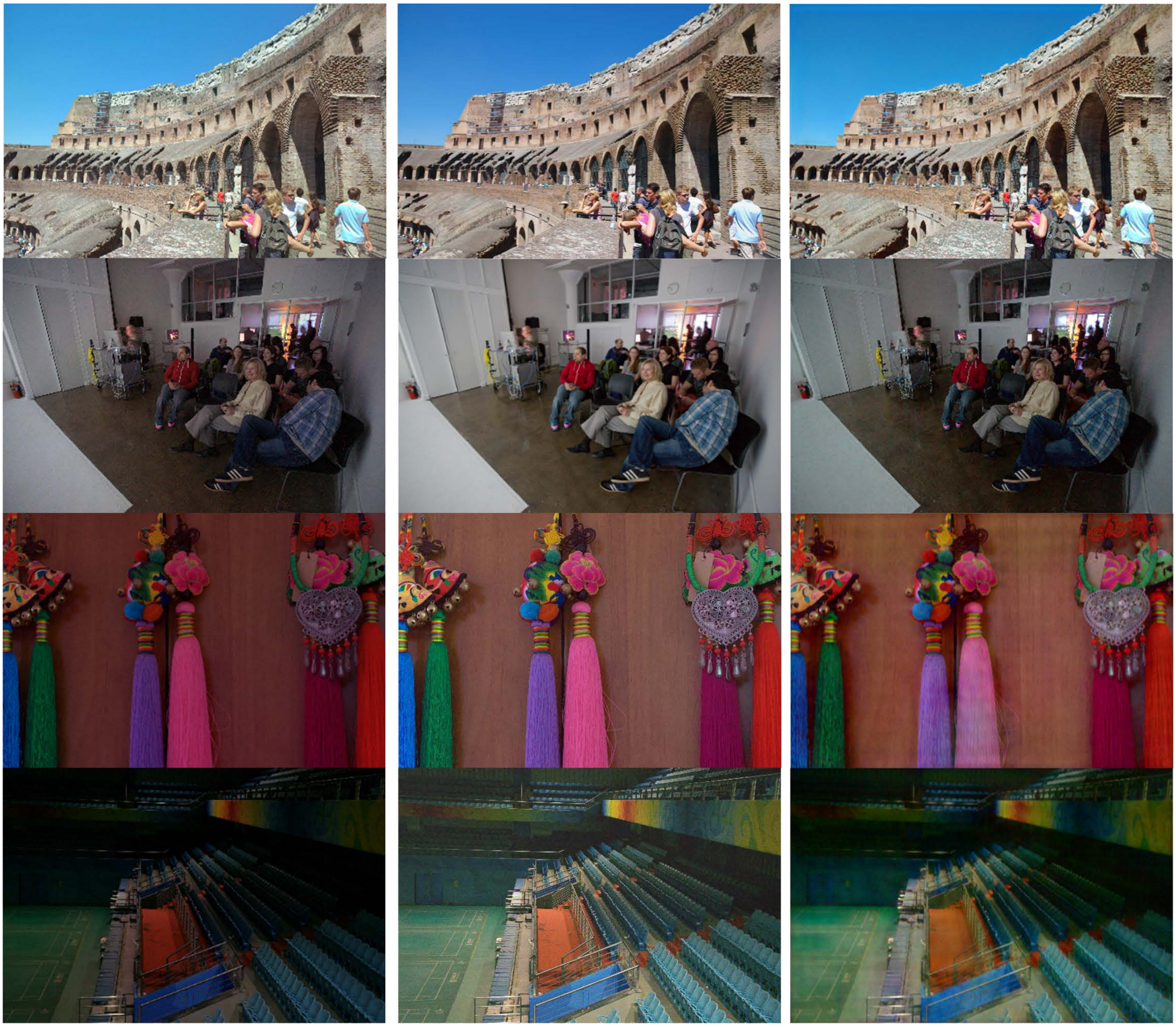

FIGURE 17. Enhance results by DeepUPE(left), DeepLPF(middle), and DSLR(right). Note that these results are enlarged versions from Figs. 7 and 8.

unpaired samples as introduced in Section III-B, however, the enhancement performance still needs to be improved to be more realistic while preserving the naturalness. To bridge the gap between supervised and unsupervised approaches, it may be helpful to take illumination properties as the learning guide into account. In addition, the problem of noise amplification needs to be effectively handled in a unified network architecture. Even though several handcrafted feature-based methods have introduced two-stage schemes (i.e., denoising + contrast enhancement) [73], [74], [75], learning-based methods, which allow for such two different tasks in a unified way, are still lacking.

- **Dataset.** As another direction, more datasets reflecting diverse real environments need to be constructed. Since low-light conditions appear unexpectedly and complicatedly in our daily pictures, such cases are probably unseen in training samples. This leads to the significant performance drop for supervised learning-based approaches. For example, extremely dark images, which are not acquired in general situations, are often under-enhanced as shown in Fig. 15. It is noteworthy that other restoration tasks, e.g., deblurring, denoising, etc., also suffer from the same issue and accordingly attempt to construct the real-world dataset, e.g., RealBlur dataset [76] for image deblurring.
- **Evaluation.** Since the good quality can be perceived differently by each person, various metrics of the visual quality assessment have been adopted for the performance evaluation as explained. Nevertheless, it is still

FIGURE 18. **Enhance results by DeepUPE(left), DeepLPF(middle), and DSLR(right). Note that these results are enlarged versions from Figs. 10, 11, 12, and 13.**

hard to reach the consensus on the performance of each method for a variety of test samples. Therefore, it is highly desirable to design the unified metric to effectively guide the development process of algorithms for low-light image enhancement.

- **Deployability.** In addition to the theoretical analysis, practical studies for deployability should not be forgotten. Mobile platforms that require reliable solutions for low-light image enhancement are rapidly increasing, thus methods to be developed need to be lightweight while keeping memory-efficient properties. A few approaches have focused on real-time operations [77], [78], however, most methods introduced in literature still require quite a large amount of parameters for restoring more visually appealing results. To be successful deployment of low-light image enhancement, considerable efforts still need to be devoted in the community of computer vision.

For the success of low-light image enhancement, challenging issues mentioned above must be carefully addressed. In addition, the combination of learning-based methods and physical modeling also needs to be considered as the future work for construction of high-performed low-light image enhancement. Even though several learning-based approaches have attempted to separately infer illumination and reflectance, the training process is not accurately and sufficiently guided due to lack of physical supports in loss functions. By incorporating the physical properties of illumination and their mathematical models into the network architecture and the training process, learning-based

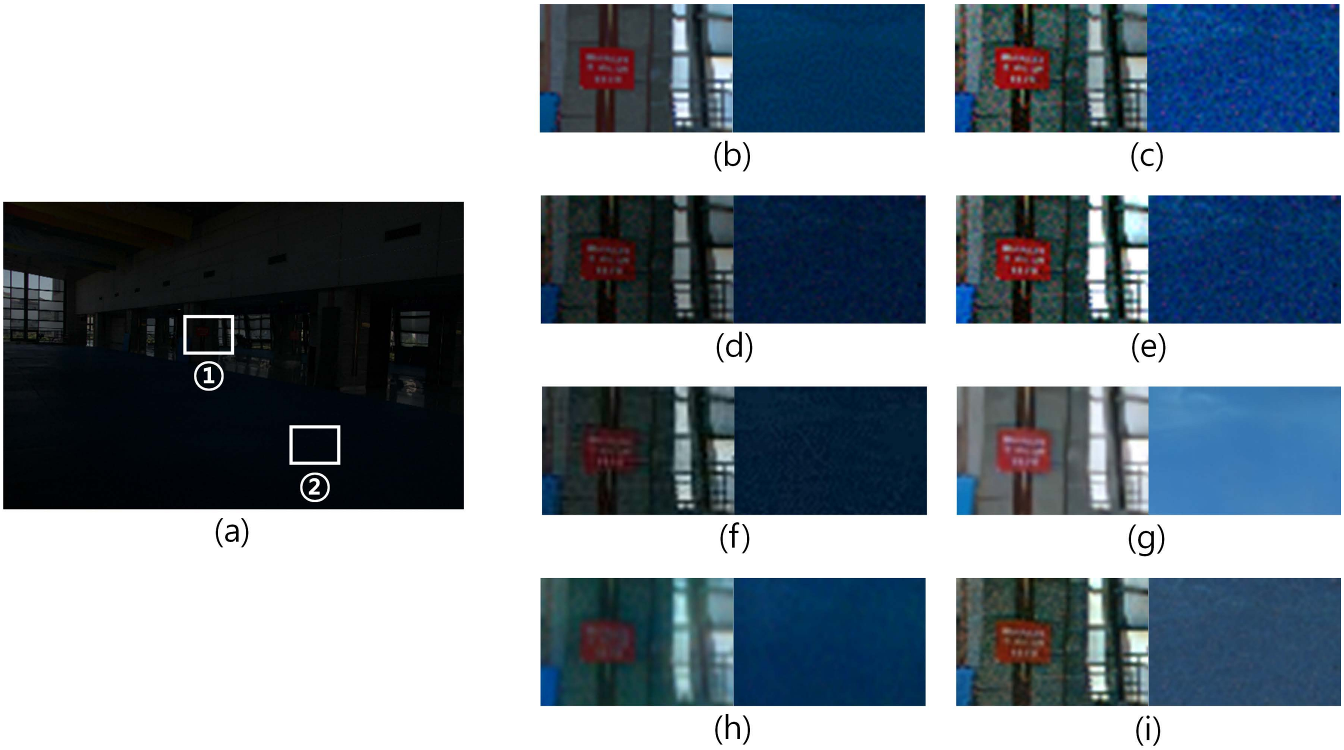

FIGURE 19. **(Enlarged version of Fig. 9) Example of the performance comparison in local regions. (a) Input selected from the LOL dataset [9]. Enlarged regions of by (b) Ground truth, (c) NPE [28], (d) SRIE [21], (e) LIME [22], (f) DeepLPF [49], (g) DRBN [39], (h) DSLR [51], and (i) EnlightenGAN [58]. Note that left and right sub-images are enhancement results for ① and ②, respectively.**

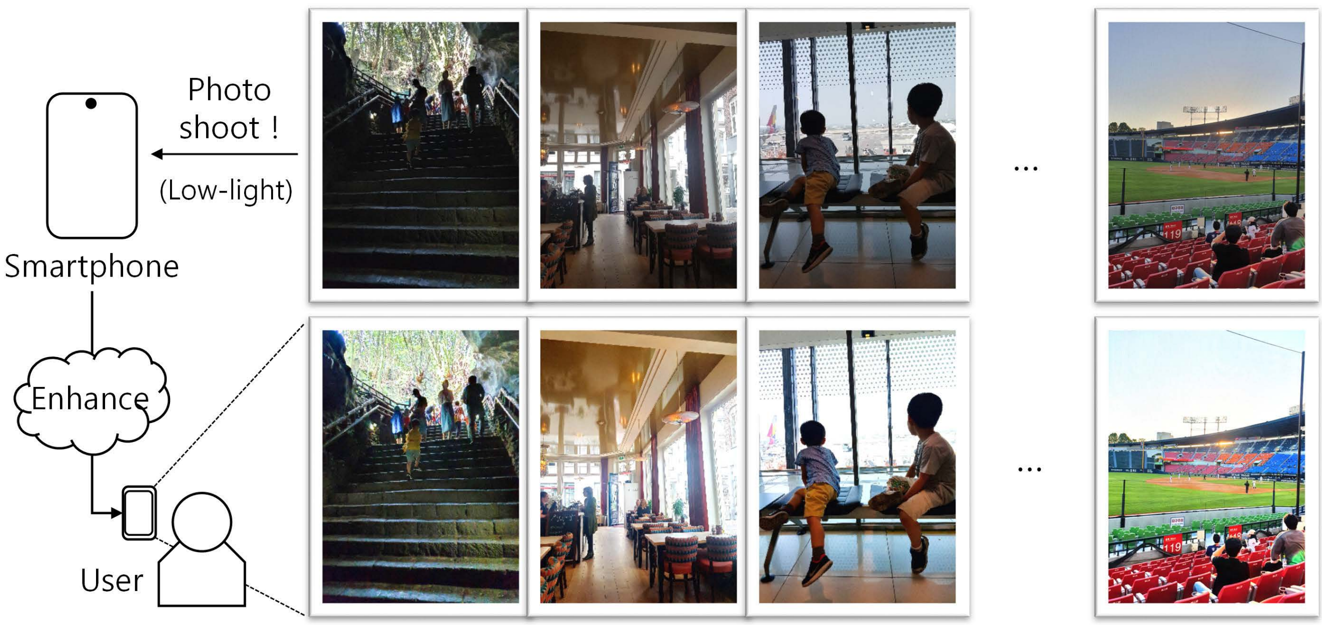

FIGURE 20. **(Enlarged version of Fig. 14) Example of the smartphone application. Note that original pictures are taken with Samsung Galaxy S10+ (top) and corresponding enhancement results are generated by DSLR [51] (bottom).**

methods are expected to provide more reliable results. Moreover, this combinations is also helpful to cope with the problem of learning bias by allowing for the general characteristic regardless of data types, which is driven from the physical model of illumination. By doing this, the technique of low-light image enhancement leads to a new generation of various applications in the field of computer vision.

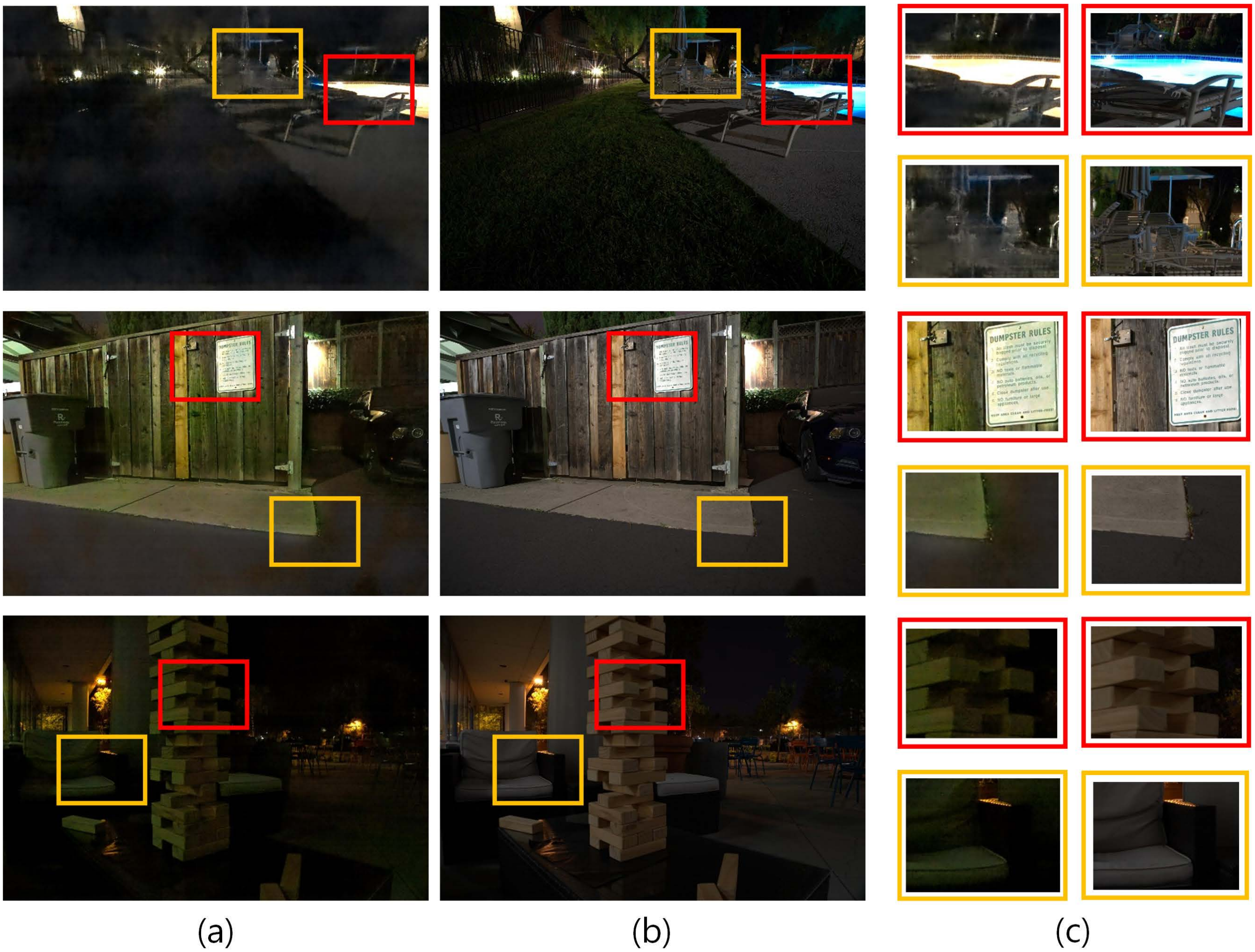

FIGURE 21. (Enlarged version of Fig. 15) Several examples of enhancement results from extremely low-light images (SID dataset [79]). (a) RED method [80]. (b) Ground truth. (c) Enlarged regions selected from (left: (a)) and (right: (b)), respectively. Note that color attributes are somewhat distorted in restored images.

V. CONCLUSION

In this review, systematic taxonomy and comprehensive analysis for low-light image enhancement have been provided. Specifically, we categorize diverse methods into two main groups, which are subsequently divided into several models according to the analysis strategy. In contrast to previous surveys, handcrafted feature-based and learning-based approaches have been reviewed in a balanced way. To give a practical guide for beginners as well as experts, both qualitative and quantitative evaluations have been conducted with various benchmark datasets. Based on experimental results, this review also provides the detailed analysis for properties of each approach based on advantages and weakness. Finally, the future direction for low-light image enhancement is also discussed in depth with consideration of deployability. We conclude that learning-based approaches are more appropriate to consider diverse conditions occurring in real-world scenarios for low-light image enhancement compared to handcrafted feature-based methods. In particular, supervised approaches via the large amount of paired training samples, e.g., DeepUPE, DeepLPF, DSLR, etc., have shown the reliable performance in general, thus can be adopted as prerequisite for various applications in the field of computer vision. Once the performance gap is overcome in the near future, unsupervised approaches will become mainstream for low-light image enhancement due to the generality nature.

APPENDIX

We provide the enlarged version of enhancement results by three representative methods, i.e., DeepUPE, DeepLPF, and DSLR, so that readers can understand limitations as well as strength of each method more clearly, and determine whether such methods are good enough for their applications or not. Please see Figs. 16, 17, 18, 19, 20, and 21, respectively.

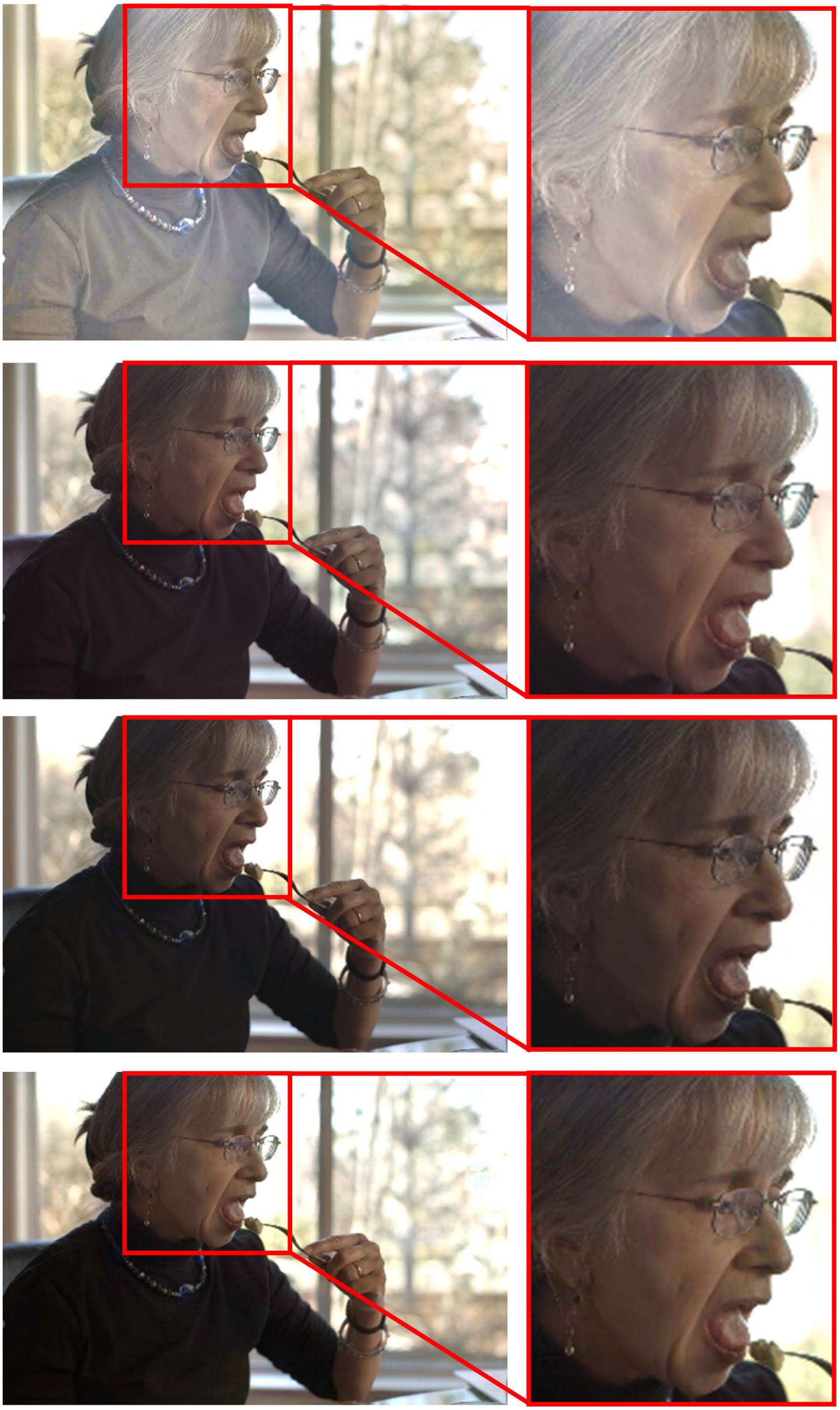

FIGURE 22. Performance comparison on the facial region from Fig. 11. From top to bottom: enhancement results by EnlightGAN [58], DeepUPE [44], DeepLPF [49], and DSLR [51].

In particular, we also provide a specific example to clearly figure out local details of enhancement results, e.g., facial region, by each method in Fig. 22.

REFERENCES

[1] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 2002.

[2] C. Guo, C. Li, J. Guo, C. C. Loy, J. Hou, S. Kwong, and R. Cong, "Zero-reference deep curve estimation for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1780–1789.

[3] Y. Guo, Y. Lu, and R. W. Liu, "Lightweight deep network-enabled real-time low-visibility enhancement for promoting vessel detection in maritime video surveillance," *J. Navigat.*, vol. 75, no. 1, pp. 230–250, Jan. 2022.

[4] W. Wang, X. Wu, X. Yuan, and Z. Gao, "An experiment-based review of low-light image enhancement methods," *IEEE Access*, vol. 8, pp. 87884–87917, 2020.

[5] C. Li, C. Guo, L.-H. Han, J. Jiang, M.-M. Cheng, J. Gu, and C. C. Loy, "Low-light image and video enhancement using deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Nov. 9, 2021, doi: 10.1109/TPAMI.2021.3126387.

[6] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[7] E. H. Land, "The Retinex theory of color vision," *Sci. Amer.*, vol. 237, no. 6, pp. 108–128, Dec. 1977.

[8] V. Bychkovsky, S. Paris, E. Chan, and F. Durand, "Learning photographic global tonal adjustment with a database of input/output image pairs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 97–104.

[9] C. Wei, W. Wang, W. Yang, and J. Liu, "Deep Retinex decomposition for low-light enhancement," in *Proc. Brit. Mach. Vis. Conf.*, 2018, pp. 1–12.

[10] A. M. Reza, "Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement," *J. VLSI signal Process. Syst. Signal, Image Video Technol.*, vol. 38, no. 1, pp. 35–44, 2004.

[11] C.-C. Sun, S.-J. Ruan, M.-C. Shie, and T.-W. Pai, "Dynamic contrast enhancement based on histogram specification," *IEEE Trans. Consum. Electron.*, vol. 51, no. 4, pp. 1300–1305, Nov. 2005.

[12] D. Menotti, L. Najman, J. Facon, and A. D. A. Araujo, "Multi-histogram equalization methods for contrast enhancement and brightness preserving," *IEEE Trans. Consum. Electron.*, vol. 53, no. 3, pp. 1186–1194, Aug. 2007.

[13] Y. Ueda and N. Suetake, "Hue-preserving color image enhancement on a vector space of convex combination coefficients," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 939–943.

[14] T. Celik and T. Tjahjadi, "Contextual and variational contrast enhancement," *IEEE Trans. Image Process.*, vol. 20, no. 12, pp. 3431–3441, Dec. 2011.

[15] C. Lee, C. Lee, and C.-S. Kim, "Contrast enhancement based on layered difference representation of 2D histograms," *IEEE Trans. Image Process.*, vol. 22, no. 12, pp. 5372–5384, Dec. 2013.

[16] K. Gu, G. Zhai, X. Yang, W. Zhang, and C. W. Chen, "Automatic contrast enhancement technology with saliency preservation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 9, pp. 1480–1494, Sep. 2015.

[17] K. Gu, G. Zhai, W. Lin, and M. Liu, "The analysis of image contrast: From quality assessment to automatic enhancement," *IEEE Trans. Cybern.*, vol. 46, no. 1, pp. 284–297, Jan. 2016.

[18] K. Gu, D. Tao, J.-F. Qiao, and W. Lin, "Learning a no-reference quality assessment model of enhanced images with big data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 4, pp. 1301–1313, Apr. 2018.

[19] R. Kimmel, M. Elad, D. Shaked, R. Keshet, and I. Sobel, "A variational framework for Retinex," *Int. J. Comput. Vis.*, vol. 52, pp. 7–23, Apr. 2003.

[20] L. I. Rudin, S. Osher, and E. Fatemi, "Nonlinear total variation based noise removal algorithms," *Phys. D, Nonlinear Phenomena*, vol. 60, nos. 1–4, pp. 259–268, 1992.

[21] X. Fu, D. Zeng, Y. Huang, X.-P. Zhang, and X. Ding, "A weighted variational model for simultaneous reflectance and illumination estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2782–2790.

[22] X. Guo, Y. Li, and H. Ling, "LIME: Low-light image enhancement via illumination map estimation," *IEEE Trans. Image Process.*, vol. 26, no. 2, pp. 982–993, Feb. 2017.

[23] M. Li, J. Liu, W. Yang, X. Sun, and Z. Guo, "Structure-revealing low-light image enhancement via robust Retinex model," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2828–2841, Jun. 2018.

[24] X. Ren, W. Yang, W.-H. Cheng, and J. Liu, "LR3M: Robust low-light enhancement via low-rank regularized Retinex model," *IEEE Trans. Image Process.*, vol. 29, pp. 5862–5876, 2020.

[25] S. Hao, X. Han, Y. Guo, X. Xu, and M. Wang, "Low-light image enhancement with semi-decoupled decomposition," *IEEE Trans. Multimedia*, vol. 22, no. 12, pp. 3025–3038, Dec. 2020.

[26] D. J. Jobson, Z.-U. Rahman, and G. A. Woodell, "Properties and performance of a center/surround Retinex," *IEEE Trans. Image Process.*, vol. 6, no. 3, pp. 451–462, Mar. 1997.

[27] D. J. Jobson, Z.-U. Rahman, and G. A. Woodell, "A multiscale Retinex for bridging the gap between color images and the human observation of scenes," *IEEE Trans. Image Process.*, vol. 6, no. 7, pp. 965–976, Jul. 1997.

[28] S. Wang, J. Zheng, H.-M. Hu, and B. Li, "Naturalness preserved enhancement algorithm for non-uniform illumination images," *IEEE Trans. Image Process.*, vol. 22, no. 9, pp. 3538–3548, Sep. 2013.

[29] Z. Liang, W. Liu, and R. Yao, "Contrast enhancement by nonlinear diffusion filtering," *IEEE Trans. Image Process.*, vol. 25, no. 2, pp. 673–686, Feb. 2016.

[30] W. Kim, R. Lee, M. Park, and S.-H. Lee, "Low-light image enhancement based on maximal diffusion values," *IEEE Access*, vol. 7, pp. 129150–129163, 2019.

[31] W. Kim, "Low-light image enhancement by diffusion pyramid with residuals," *J. Vis. Commun. Image Represent.*, vol. 81, Nov. 2021, Art. no. 103364.

[32] J. Stander, R. Mech, and J. Ostermann, "Detection of moving cast shadows for object segmentation," *IEEE Trans. Multimedia*, vol. 1, no. 1, pp. 65–76, Mar. 1999.

[33] W. Zhang, X. Zhong Fang, and Y. Xu, "Detection of moving cast shadows using image orthogonal transform," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, 2006, pp. 626–629.

[34] W. Kim, S. Suh, W. Hwang, and J.-J. Han, "SVD face: Illumination-invariant face representation," *IEEE Signal Process. Lett.*, vol. 21, no. 11, pp. 1336–1340, Nov. 2014.

[35] X. Tan and B. Triggs, "Enhanced local texture feature sets for face recognition under different lighting conditions," *IEEE Trans. Image Process.*, vol. 19, no. 6, pp. 1635–1650, Jun. 2010.

[36] W. Kim, "Image enhancement using patch-based principal energy analysis," *IEEE Access*, vol. 6, pp. 72620–72628, 2018.

[37] A. Ignatov, N. Kobyshev, R. Timofte, and K. Vanhoey, "DSLR-quality photos on mobile devices with deep convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3297–3305.

[38] F. Lv, F. Lu, J. Wu, and C. Lim, "MBLLEN: Low-light image/video enhancement using CNNs," in *Proc. Brit. Mach. Vis. Conf.*, Sep. 2018, pp. 1–13.

[39] W. Yang, S. Wang, Y. Fang, Y. Wang, and J. Liu, "From fidelity to perceptual quality: A semi-supervised approach for low-light image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3060–3069.

[40] L.-W. Wang, Z.-S. Liu, W.-C. Siu, and D. P. Lun, "Lightening network for low-light image enhancement," *IEEE Trans. Image Process.*, vol. 29, pp. 7984–7996, 2020.

[41] J. Cai, S. Gu, and L. Zhang, "Learning a deep single image contrast enhancer from multi-exposure images," *IEEE Trans. Image Process.*, vol. 27, no. 4, pp. 2049–2062, Apr. 2018.

[42] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, Oct. 2015, pp. 234–241.

[43] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[44] R. Wang, Q. Zhang, C.-W. Fu, X. Shen, W.-S. Zheng, and J. Jia, "Underexposed photo enhancement using deep illumination estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6842–6850.

[45] Y. Zhang, J. Zhang, and X. Guo, "Kindling the darkness: A practical low-light image enhancer," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 1632–1640.

[46] K. Xu, X. Yang, B. Yin, and R. W. H. Lau, "Learning to restore low-light images via decomposition-and-enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2281–2290.

[47] W. Yang, W. Wang, H. Huang, S. Wang, and J. Liu, "Sparse gradient regularized deep Retinex network for robust low-light image enhancement," *IEEE Trans. Image Process.*, vol. 30, pp. 2072–2086, 2021.

[48] M. Gharbi, J. Chen, J. T. Barron, S. W. Hasinoff, and F. Durand, "Deep bilateral learning for real-time image enhancement," *ACM Trans. Graph.*, vol. 36, no. 4, pp. 1–12, Jul. 2017.

[49] S. Moran, P. Marza, S. McDonagh, S. Parisot, and G. Slabaugh, "DeepLPF: Deep local parametric filters for image enhancement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12826–12835.
[50] J. Li, J. Li, F. Fang, F. Li, and G. Zhang, "Luminance-aware pyramid network for low-light image enhancement," *IEEE Trans. Multimedia*, vol. 23, pp. 3153–3165, 2021.
[51] S. Lim and W. Kim, "DSLR: Deep stacked Laplacian restorer for low-light image enhancement," *IEEE Trans. Multimedia*, vol. 23, pp. 4272–4284, 2021.
[52] Y. Lu, Y. Guo, R. W. Liu, and W. Ren, "MTRBNet: Multi-branch topology residual block-based network for low-light enhancement," *IEEE Signal Process. Lett.*, vol. 29, pp. 1127–1131, 2022.
[53] Y.-S. Chen, Y.-C. Wang, M.-H. Kao, and Y.-Y. Chuang, "Deep photo enhancer: Unpaired learning for image enhancement from photographs with GANs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6306–6314.
[54] J. Park, J.-Y. Lee, D. Yoo, and I. S. Kweon, "Distort-and-recover: Color enhancement using deep reinforcement learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5928–5936.
[55] S. Kosugi and T. Yamasaki, "Unpaired image enhancement featuring reinforcement-learning-controlled image editing software," in *Proc. AAAI Conf. Atrifi. Intell.*, Apr. 2020, pp. 11296–11303.
[56] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, Jul. 2017, pp. 214–223.
[57] Z. Ni, W. Yang, S. Wang, L. Ma, and S. Kwong, "Unpaired image enhancement with quality-attention generative adversarial network," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 1697–1705.
[58] Y. Jiang, X. Gong, D. Liu, Y. Cheng, and C. Fang, "EnlightenGAN: Deep light enhancement without paired supervision," *IEEE Trans. Image Process.*, vol. 30, pp. 2340–2349, 2021.
[59] C. Li, C. Guo, and C. L. Chen, "Learning to enhance low-light image via zero-reference deep curve estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 8, pp. 4225–4238, Aug. 2022, doi: 10.1109/TPAMI.2021.3063604.
[60] A. Zhu, L. Zhang, Y. Shen, Y. Ma, S. Zhao, and Y. Zhou, "Zero-shot restoration of underexposed images via robust Retinex decomposition," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2020, pp. 1–6.
[61] H.-U. Kim, Y. J. Koh, and C.-S. Kim, "Global and local enhancement networks for paired and unpaired image enhancement," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 339–354.
[62] W. Yang, S. Wang, Y. Fang, Y. Wang, and J. Liu, "Band representation-based semi-supervised low-light image enhancement: Bridging the gap between signal fidelity and perceptual quality," *IEEE Trans. Image Process.*, vol. 30, pp. 3461–3473, 2021.
[63] NASA. *Retinex Image Processing*. Accessed: Jul. 2019. [Online]. Available: https://dragon.larc.nasa.gov/retinex/pao/news/
[64] P. Sen, N. K. Kalantari, M. Yaesoubi, S. Darabi, B. G. Dan, and E. Shechtman, "Robust patch-based HDR reconstruction of dynamic scenes," *ACM Trans. Graph.*, vol. 31, no. 6, pp. 1–11, Nov. 2012.
[65] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a completely blind image quality analyzer," *IEEE Sign. Process. Lett.*, vol. 20, no. 3, pp. 209–212, Mar. 2013.
[66] K. Gu, S. Wang, G. Zhai, S. Ma, X. Yang, W. Lin, W. Zhang, and W. Gao, "Blind quality assessment of tone-mapped images via analysis of information, naturalness, and structure," *IEEE Trans. Multimedia*, vol. 18, no. 3, pp. 432–443, Mar. 2016.
[67] K. Gu, W. Lin, G. Zhai, X. Yang, W. Zhang, and C. W. Chen, "No-reference quality metric of contrast-distorted images based on information maximization," *IEEE Trans. Cybern.*, vol. 47, no. 12, pp. 4559–4565, Dec. 2017.
[68] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.
[69] J. Wu, J. Ma, F. Liang, W. Dong, G. Shi, and W. Lin, "End-to-end blind image quality prediction with cascaded deep neural network," *IEEE Trans. Image Process.*, vol. 29, pp. 7414–7426, 2020.
[70] Y. Yuan, W. Yang, W. Ren, J. Liu, W. J. Scheirer, and W. Zhangyang, "UG^{2+} track 2: A collective benchmark effort for evaluating and advancing image understanding in poor visibility environments," 2019, *arXiv:1904.04474*.
[71] Y.-T. Huang, Y.-T. Peng, and W.-H. Liao, "Enhancing object detection in the dark using U-Net based restoration module," in *Proc. 16th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Sep. 2019, pp. 1–8.
[72] Y. P. Loh and C. S. Chan, "Getting to know low-light images with the exclusively dark dataset," *Comput. Vis. Image Understand.*, vol. 178, pp. 30–42, Jan. 2019.
[73] L. Li, R. Wang, W. Wang, and W. Gao, "A low-light image enhancement method for both denoising and contrast enlarging," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 3730–3734.
[74] J. Lim, J.-H. Kim, J.-Y. Sim, and C.-S. Kim, "Robust contrast enhancement of noisy low-light images: Denoising-enhancement-completion," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2015, pp. 4131–4135.
[75] H. Su and C. Jung, "Low light image enhancement based on two-step noise suppression," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2017, pp. 1977–1981.
[76] J. Rim, H. Lee, J. Won, and S. Cho, "Real-world blur dataset for learning and benchmarking deblurring algorithms," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 184–201.
[77] M. Lamba, A. Balaji, and K. Mitra, "Towards fast and light-weight restoration of dark images," in *Proc. Brit. Mach. Vis. Conf.*, Sep. 2018, pp. 1–14.
[78] S. Gu, Y. Li, L. Van Gool, and R. Timofte, "Self-guided network for fast image denoising," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2511–2520.
[79] C. Chen, Q. Chen, J. Xu, and V. Koltun, "Learning to see in the dark," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3291–3300.
[80] M. Lamba and K. Mitra, "Restoring extremely dark images in real time," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3487–3497.

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